Flight_Booking_Prediction

A Project Report

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By

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Business Context

An aviation company offering both domestic and international travel services aims to improve its customer acquisition strategy in an increasingly competitive **digital marketing environment**. Traditionally, the company has relied on broad-based outreach methods such as telecalling and generic advertisements. However, these approaches are **costly, less targeted, and yield low conversion rates**.

Industry benchmarks indicate that **average digital marketing conversion rates typically range between 2% and 5%**, and in the travel and aviation sector, conversion rates are often **lower due to high competition, multiple alternatives, and price-sensitive customers**. This means that a large proportion of marketing expenditure is spent on users who do not eventually purchase tickets.

To address this challenge, the company has partnered with a social networking platform to leverage **customer digital behavior and social engagement data** for targeted advertising. Since digital advertising platforms charge based on impressions, clicks, or engagement, **untargeted campaigns significantly increase customer acquisition costs**. Therefore, it has become essential for the company to focus its marketing efforts only on customers who are **most likely to convert**, thereby improving conversion efficiency and return on investment.

Objective of the Project

As a data scientist, the objective is to build **predictive classification models** that estimate the **propensity of customers to purchase airline tickets** based on their online behavior and engagement patterns available in the dataset.

Given that **device usage strongly influences browsing and purchasing behavior**, two separate predictive models are developed:

- One for **laptop users**
- One for **mobile users**

These models aim to identify high-probability customers who are more likely to purchase tickets, enabling the company to:

- Reduce unnecessary digital advertising spend
- Improve effective conversion rates
- Run more efficient and personalized marketing campaigns

Accurate prediction models will help the company **maximize conversions while minimizing marketing costs**, ultimately leading to improved business performance and profitability.

Data Description:

Column Name	Description
UserID	Unique ID of user
Buy_ticket	Buy a ticket next month
Yearly_avg_view_on_travel_page	Average yearly views on any travel-related page by user
preferred_device	Through which device user prefer to do login
total_likes_on_outstation_checkin_given	Total number of likes given by a user on out-of-station check-ins in the last year
yearly_avg_Outstation_checkins	Average number of out-of-station check-ins done by the user
member_in_family	Total number of relationships mentioned by the user in the account
preferred_location_type	Preferred type of location for travelling of user
Yearly_avg_comment_on_travel_page	Average yearly comments on any travel-related page by user

total_likes_on_outofstation_checkin_received	Total number of likes received by a user on out-of-station check-ins in the last year
week_since_last_outstation_checkin	Number of weeks since last out-of-station check-in update by user
following_company_page	Whether the customer is following the company page (Yes or No)
montly_avg_comment_on_company_page	Average monthly comments on the company page by user
working_flag	Whether the customer is working or not
travelling_network_rating	Does the user have close friends who also like travelling? 1 is high and 4 is lowest
Adult_flag	Whether the customer is an adult or not
Daily_Avg_mins_spend_on_traveling_page	Average time spent on the company page by user on a daily basis

Table 1: Data Description

Relevance of Conversion Rate to This Project

The dataset used in this project contains customer demographic details, device preferences, and digital behavior indicators, along with a target variable representing whether a customer purchased an airline ticket or not. Since only a small proportion of digitally engaged users actually convert into buyers, predicting purchase likelihood becomes critical. By identifying customers who are more likely to convert, the company can improve its effective conversion rate without increasing marketing spend.

Scope of the Project

The scope of this project is limited to analyzing the provided customer dataset to identify patterns in digital behavior and predict ticket purchase propensity. The study includes exploratory data analysis, data preprocessing, and development of classification models for mobile and laptop users. Model performance is evaluated using standard classification metrics.

Beneficiaries of the Project

This project benefits multiple stakeholders:

- **Business / Aviation Company:**
  Reduced marketing expenditure, improved conversion rates, and better return on investment by targeting only high-probability customers.
- **Marketing Teams:**
  Data-driven insights that help design focused and personalized digital campaigns instead of mass marketing.
- **Customers:**
  Improved user experience by receiving relevant and personalized advertisements rather than generic promotional messages.

Significance of the Project

This project helps the aviation company move from broad-based marketing to data-driven targeted advertising. By identifying high-probability customers, the company can optimize marketing spend, improve conversion efficiency, and enhance customer experience. The study also demonstrates the practical application of machine learning in solving real-world digital marketing challenges.

Data Overview

1. Import the required packages.
2. Loaded the dataset flight_customer_data.xlsx using **read_csv()**.
3. Used the **head()** to get the first five rows of the data.

```
df.head()
```

	UserID	Taken_product	Yearly_avg_view_on_travel_page	preferred_device	total_likes_on_outstation_checkin_given	yearly_avg_Outstation_checkins	member_in_family	preferred_location_type	Yearly_avg_comment_on_travel_pag
0	1000001	Yes	307.0	iOS and Android	38570.0	1	2	Financial	94
1	1000002	No	367.0	iOS	9765.0	1	1	Financial	61
2	1000003	Yes	277.0	iOS and Android	48055.0	1	2	Other	92
3	1000004	No	247.0	iOS	48720.0	1	4	Financial	56
4	1000005	No	202.0	iOS and Android	20685.0	1	1	Medical	40

Next steps: Generate code with df New interactive sheet

Table 2:First Five Rows Of DataFrame

4. Used the **tail()** to get the last five rows of the data.

```
df.tail()
```

	UserID	Taken_product	Yearly_avg_view_on_travel_page	preferred_device	total_likes_on_outstation_checkin_given	yearly_avg_Outstation_checkins	member_in_family	preferred_location_type	Yearly_avg_comment_on_travel
11755	1011756	No	279.0	Laptop	30987.0	23	2	Historical site	
11756	1011757	No	305.0	Tab	21510.0	6	1	Historical site	
11757	1011758	No	214.0	Tab	5478.0	4	3	Beach	
11758	1011759	No	382.0	Laptop	35851.0	2	3	Historical site	
11759	1011760	No	270.0	Tab	22025.0	8	3	Historical site	

Table 3: Last Five Rows Of DataFrame

5. Checked the shape of the dataset using the **shape method.**
    a. The dataset has 11760 rows and 17 columns.
6. Checked the datatype of all the columns in the dataset using info().
    a. There are 17 columns among which 7 of these columns are integer datatypes, 7 of them being objects and 3 of them being a float.
    b. There are some missing values in a few columns in the dataset which need to be treated.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11760 entries, 0 to 11759
Data columns (total 17 columns):
 #   Column                                      Non-Null Count  Dtype
---  ------                                      --------------  -----
 0   UserID                                      11760 non-null  int64
 1   Taken_product                               11760 non-null  object
 2   Yearly_avg_view_on_travel_page              11179 non-null  float64
 3   preferred_device                            11707 non-null  object
 4   total_likes_on_outstation_checkin_given     11379 non-null  float64
 5   yearly_avg_Outstation_checkins              11685 non-null  object
 6   member_in_family                            11760 non-null  object
 7   preferred_location_type                     11729 non-null  object
 8   Yearly_avg_comment_on_travel_page           11554 non-null  float64
 9   total_likes_on_outofstation_checkin_received 11760 non-null  int64
 10  week_since_last_outstation_checkin          11760 non-null  int64
 11  following_company_page                      11657 non-null  object
 12  montly_avg_comment_on_company_page          11760 non-null  int64
 13  working_flag                                11760 non-null  object
 14  travelling_network_rating                   11760 non-null  int64
 15  Adult_flag                                  11760 non-null  int64
 16  Daily_Avg_mins_spend_on_traveling_page      11760 non-null  int64
dtypes: float64(3), int64(7), object(7)
memory usage: 1.5+ MB
```

Table 4: Data Frame Information

7. There are no duplicate rows present in the dataframe checked using duplicated().
8. There are some missing values present in the dataframe checked using **isnull().**
    a. Handled all numeric missing values by replacing it with the median value.
    b. Handled all categorical missing values by replacing it with mode value.

```
# checking for missing values
df.isnull().sum()
```

	0
UserID	0
Taken_product	0
Yearly_avg_view_on_travel_page	581
preferred_device	53
total_likes_on_outstation_checkin_given	381
yearly_avg_Outstation_checkins	75
member_in_family	0
preferred_location_type	31
Yearly_avg_comment_on_travel_page	206
total_likes_on_outofstation_checkin_received	0
week_since_last_outstation_checkin	0
following_company_page	103
montly_avg_comment_on_company_page	0
working_flag	0
travelling_network_rating	0
Adult_flag	0
Daily_Avg_mins_spend_on_traveling_page	0

dtype: int64

```
df.isnull().sum()
```

	0
UserID	0
Taken_product	0
Yearly_avg_view_on_travel_page	0
preferred_device	0
total_likes_on_outstation_checkin_given	0
yearly_avg_Outstation_checkins	0
member_in_family	0
preferred_location_type	0
Yearly_avg_comment_on_travel_page	0
total_likes_on_outofstation_checkin_received	0
week_since_last_outstation_checkin	0
following_company_page	0
montly_avg_comment_on_company_page	0
working_flag	0
travelling_network_rating	0
Adult_flag	0
Daily_Avg_mins_spend_on_traveling_page	0

dtype: int64

Table 5: Before and After Handling Null Values

9. Checked the statistical description for the dataset

```
df.describe(include='all').T
# setting include='all' will get the statistical summary for both the numerical and categorical variables.
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
UserID	11760.0	NaN	NaN	NaN	1005880.5	3394.963917	1000001.0	1002940.75	1005880.5	1008820.25	1011760.0
Taken_product	11760	2	No	9864	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Yearly_avg_view_on_travel_page	11760.0	NaN	NaN	NaN	280.345153	66.51133	35.0	233.0	271.0	322.0	464.0
preferred_device	11760	10	Tab	4225	NaN	NaN	NaN	NaN	NaN	NaN	NaN
total_likes_on_outstation_checkin_given	11760.0	NaN	NaN	NaN	28167.420748	14150.080463	3570.0	16697.25	28076.0	40115.25	252430.0
yearly_avg_Outstation_checkins	11760.0	NaN	NaN	NaN	8.196259	8.65095	1.0	1.0	4.0	14.0	29.0
member_in_family	11760.0	7.0	3.0	4561.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
preferred_location_type	11760	15	Beach	2455	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Yearly_avg_comment_on_travel_page	11760.0	NaN	NaN	NaN	74.793707	23.81528	3.0	57.0	75.0	92.0	815.0
total_likes_on_outofstation_checkin_received	11760.0	NaN	NaN	NaN	6531.699065	4706.613785	1009.0	2940.75	4948.0	8393.25	20065.0
week_since_last_outstation_checkin	11760.0	NaN	NaN	NaN	3.203571	2.616365	0.0	1.0	3.0	5.0	11.0
following_company_page	11760	4	No	8458	NaN	NaN	NaN	NaN	NaN	NaN	NaN
montly_avg_comment_on_company_page	11760.0	NaN	NaN	NaN	28.661565	48.660504	11.0	17.0	22.0	27.0	500.0
working_flag	11760	2	No	9952	NaN	NaN	NaN	NaN	NaN	NaN	NaN
travelling_network_rating	11760.0	NaN	NaN	NaN	2.712245	1.080887	1.0	2.0	3.0	4.0	4.0
Adult_flag	11760.0	NaN	NaN	NaN	0.793878	0.851823	0.0	0.0	1.0	1.0	3.0
Daily_Avg_mins_spend_on_traveling_page	11760.0	NaN	NaN	NaN	13.817432	9.070657	0.0	8.0	12.0	18.0	270.0

Table 6: Data Description Table

- The dataset has around 11,760 records, with a few columns containing missing values.
- Most users have not taken the product, showing low conversion.
- The beach is the most preferred travel location among users.
- Tab is the most commonly used device for browsing travel pages.
- User engagement (likes and comments) shows high variation, indicating the presence of outliers.
- Most users belong to small families with around 3 members.
- Users spend on average about 12 minutes per day on travel pages.
- A small group of users is highly active, contributing to very high maximum values in engagement.

Device-Wise Segregation

Categorical variables such as **following_company_page** and **taken_product** were converted into binary numerical values (Yes = 1, No = 0) to make them suitable for machine learning models.
Customer behavior varies significantly based on the device used. Since this project aims to build separate predictive models for mobile and laptop users, device-wise segregation was performed.
The preferred device variable contained multiple inconsistent values, which were standardized into three categories: Mobile, Laptop, and Other and stored to device_group. After cleaning, the dataset consisted of **6,278 mobile users**, **5,333 laptop users**, 149 users under other devices. Due to the very small size of the "Other" category, it was excluded from further analysis.
Separate datasets were created for mobile and laptop users to capture device-specific behavior patterns, enabling more accurate prediction of ticket purchase likelihood and more efficient targeted marketing.

```
df['device_group'].value_counts()
```

device_group	count
Mobile	6278
Laptop	5333
Other	149

dtype: int64

```
df_mobile = df_device[df_device['device_group'] == 'Mobile']
df_laptop = df_device[df_device['device_group'] == 'Laptop']
```

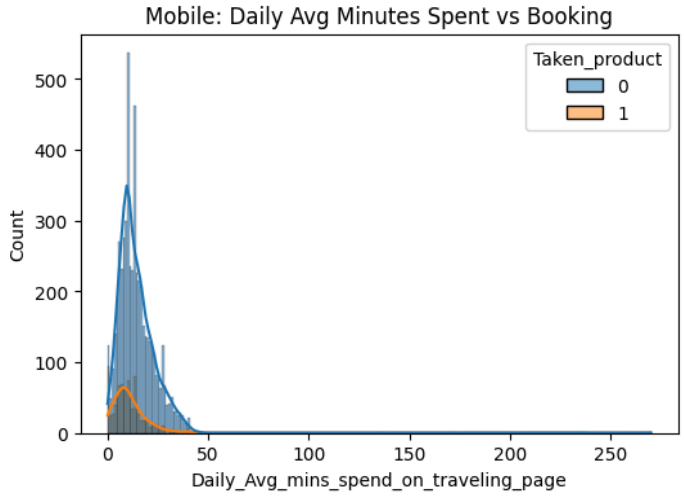
Fig 1: Device-Wise Segregation

Exploratory Data Analysis

Univariate Analysis

Daily Avg Minutes Spent On Travelling Page vs Booking For Mobile

Fig 2: Mobile- Daily Avg Minutes Spent vs Booking



Observations:

1. Most non-booked mobile users spend more time on the traveling page.
2. Booked users usually spend less time and make quicker decisions.
3. The distribution is right-skewed, especially for non-booked users.
4. Higher time spent on mobile does not guarantee booking.
5. Longer sessions likely indicate browsing or confusion rather than intent.

Daily Avg Minutes Spent On Travelling Page vs Booking For Laptop

Observations:

1. Laptop users who did not book tend to spend more time on the traveling page.
2. Users who booked tickets generally spend less and more consistent time.
3. The distribution shows less extreme skewness compared to mobile users.
4. Longer time spent on a laptop does not strongly increase booking probability.
5. Laptop users appear more focused, completing bookings in shorter sessions.

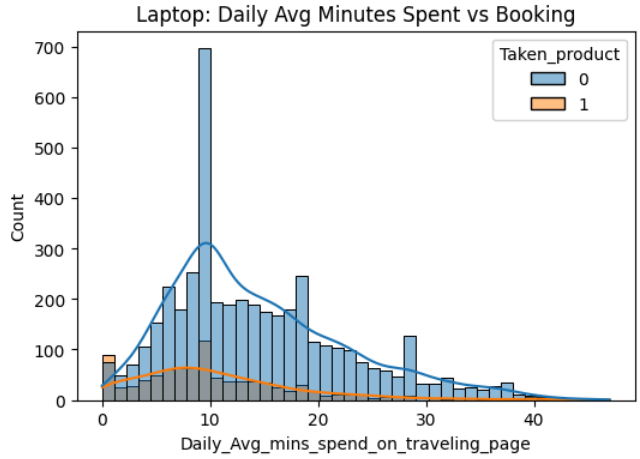


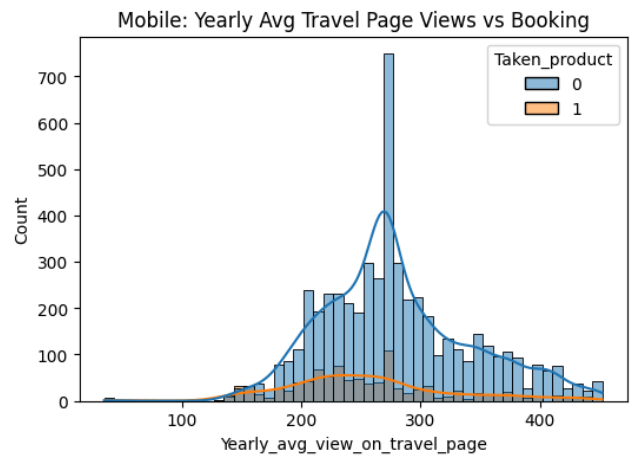
Fig 3: Laptop-: Daily Avg Minutes Spent vs Booking

Yearly Avg On :Travel Page Views vs Booking For Mobile

Observations:

1. Most mobile users with high yearly page views still do not book tickets.
2. Booked users are concentrated around moderate page views, not extremely high values.
3. The distribution is right-skewed, especially for non-booked users.
4. Higher page views on mobile do not directly lead to booking.
5. Repeated viewing likely reflects searching or comparison behavior rather than purchase intent.

Fig 4: Mobile-Yearly Avg Minutes Spent vs Booking



Yearly Avg On :Travel Page Views vs Booking For Laptop

Observations:

1. Many laptop users view travel pages frequently but do not complete booking.
2. Booked users are mostly concentrated around moderate page view counts.
3. The distribution is less skewed compared to mobile users.
4. Extremely high page views do not strongly improve booking chances.
5. Laptop users with moderate viewing behavior show higher booking intent.

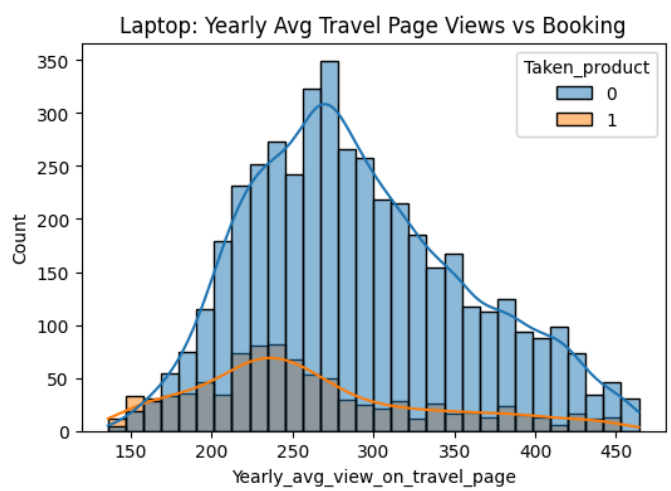


Fig 5: Laptop-Yearly Avg Minutes Spent vs Booking

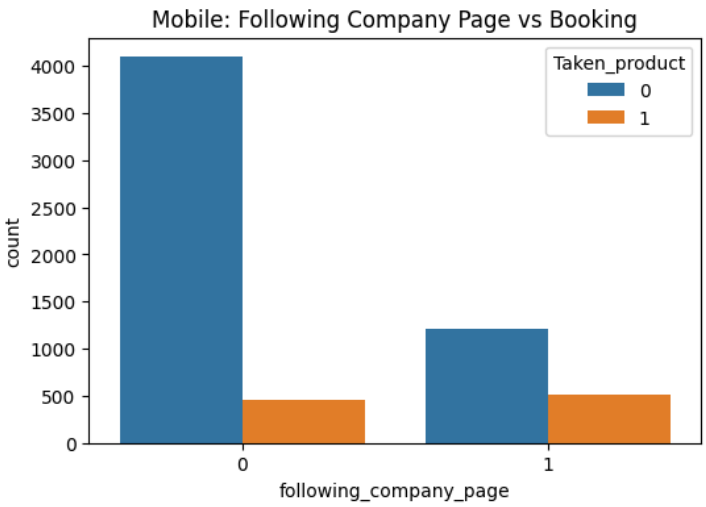
Bivariant Analysis

Following Company Page vs Booking For Mobile

Observations:

1. Most mobile users who do not follow the company page do not book tickets.
2. A higher proportion of bookings comes from users who follow the company page.
3. Following the company page shows a positive association with booking.
4. Non-followers have lower engagement and conversion.
5. Social media following helps improve trust and booking intent.

Fig 6: Mobile - Following Company Page vs Booking

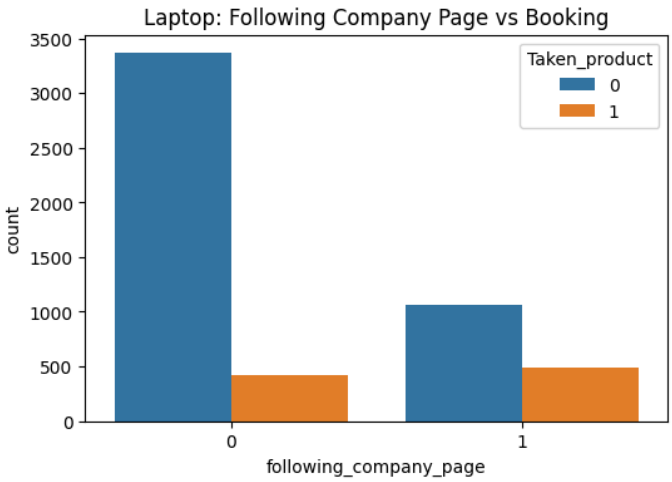


Following Company Page vs Booking For Laptop

Observations:

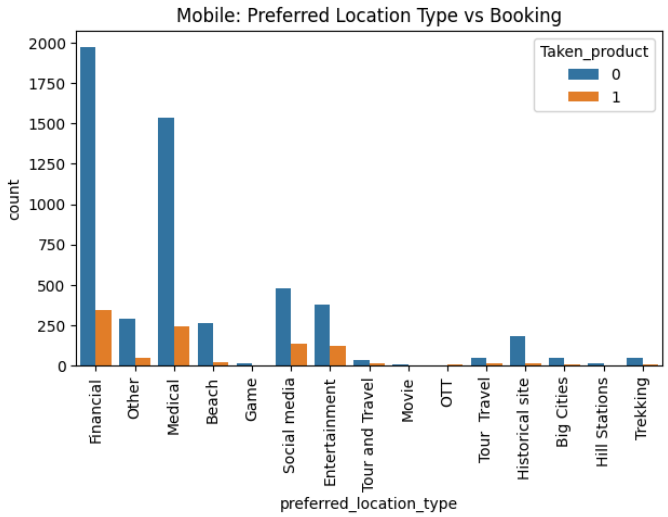
1. Most laptop users who do not follow the company page do not book tickets.
2. Users who follow the company page show a higher booking count.
3. Following the company page is positively associated with ticket booking.
4. Laptop users who follow the page appear more engaged and confident.
5. Social media presence helps increase trust and conversion on laptops.

Fig 7: Laptop - Following Company Page vs Booking



Preferred Location Type vs Booking For Mobile

Fig 8: Mobile - Preferred Location Type Vs Booking

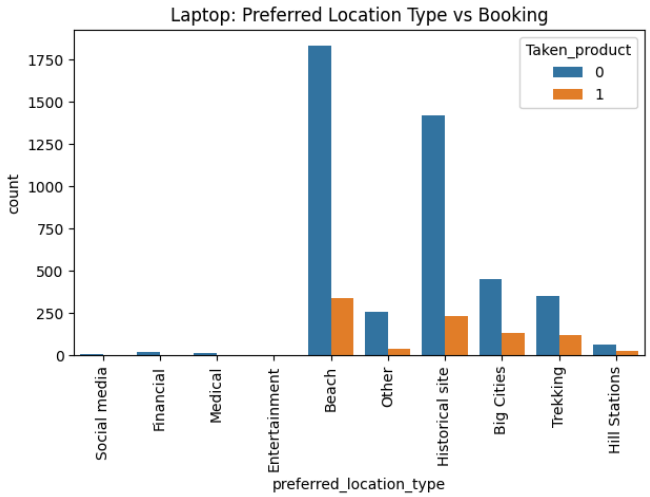


Observations:

1. Most mobile users who prefer Financial and Medical locations do not book tickets.
2. Higher bookings are observed for users interested in Social media and Entertainment-related locations.
3. Categories like (Movie, OTT, Game) have very low booking volumes.

Preferred Location Type vs Booking For Laptop

Fig 9:Laptop - Preferred Location Type Vs Booking



Observations:

1. Most laptop users preferring Beach and Historical sites do not book tickets.
2. Bookings are relatively higher for Beach locations compared to other categories.
3. Trekking shows a more balanced pattern between booked and not booked users.
4. Big Cities and Hill Stations have lower overall booking volumes.

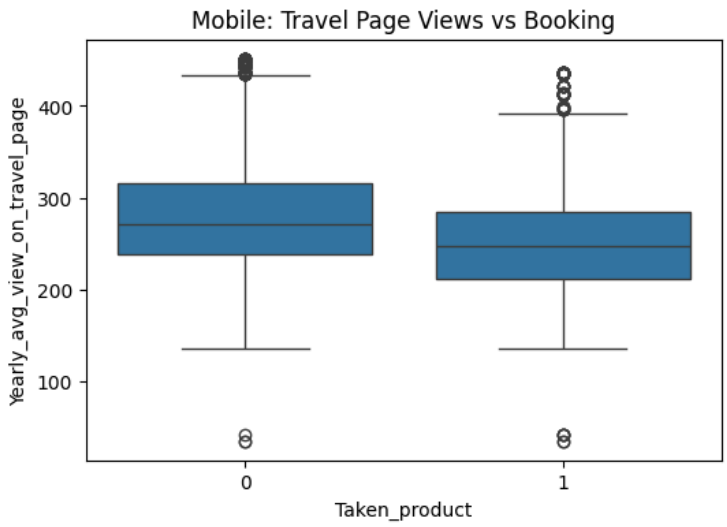
Multivariate Analysis

Yearly Average Page Views vs Booking For Mobile

Observations:

1. Non-booked users generally have a slightly higher median travel page views.
2. Booked users show lower median page views, indicating quicker decisions.
3. Both groups have wide spread, showing varied browsing behavior.
4. Presence of outliers suggests some users view pages excessively without booking.
5. Higher page views on mobile do not guarantee booking.

Fig 10: Mobile - Travel Page Views vs Booking

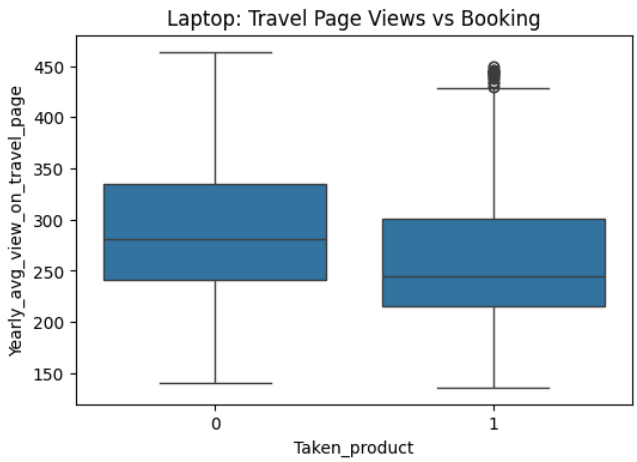


Yearly Average Page Views vs Booking For Laptop

Observations:

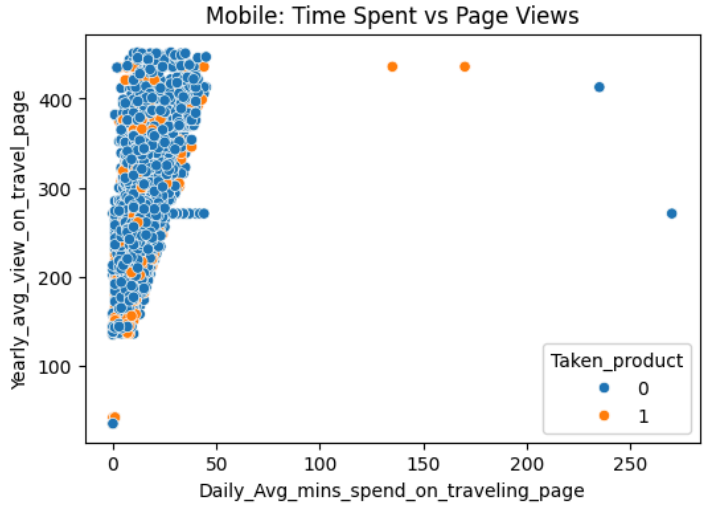
1. Non-booked laptop users have a higher median travel page views.
2. Booked users show lower and more consistent page views.
3. The spread of page views is narrower for booked users, indicating focused browsing.
4. Presence of outliers suggests some users browse extensively without booking.
5. Higher page views on laptop do not directly lead to booking.

Fig 11: Laptop - Travel Page Views vs Booking



Time Spent vs Views For Mobile

Fig 12: Mobile - Time Spent vs Views For Mobile

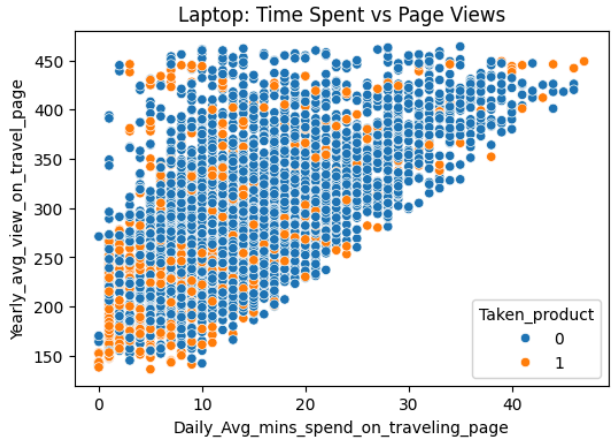


Observations:

1. Non-booked mobile users dominate the plot and are spread across both high time spent and high page views.
2. Booked users are few and mostly appear at low to moderate time spent.
3. There is a positive relationship between time spent and page views for mobile users.
4. Several users spend a lot of time with high page views but still do not book.
5. This indicates that on mobile, long browsing does not translate into booking.

Time Spent On Travel Pages vs Views For Laptop

Fig 13: Laptop - Time Spent vs Views For Mobile



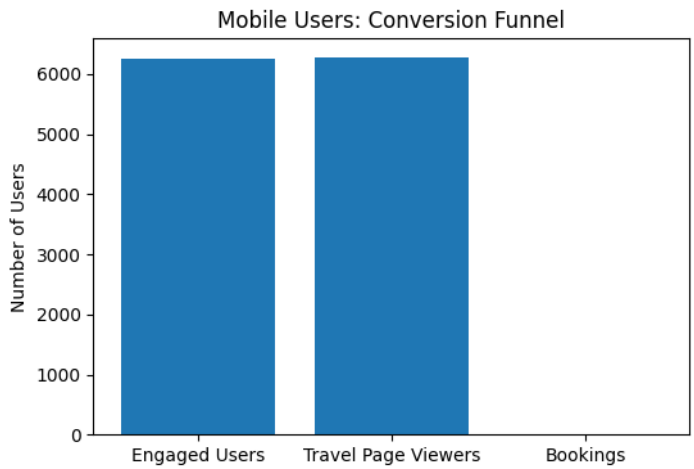
Observations:

1. Non-booked laptop users are spread across higher time spent and higher page views.
2. Booked users are mostly concentrated at moderate time spent and page views.
3. There is a clear positive relationship between time spent and page views.
4. Extremely high browsing does not necessarily lead to booking.
5. Laptop users who book appear more focused, completing decisions with moderate engagement.

Funnel For Mobile Users

Observations:

Fig 14: Mobile-Conversion Funnel

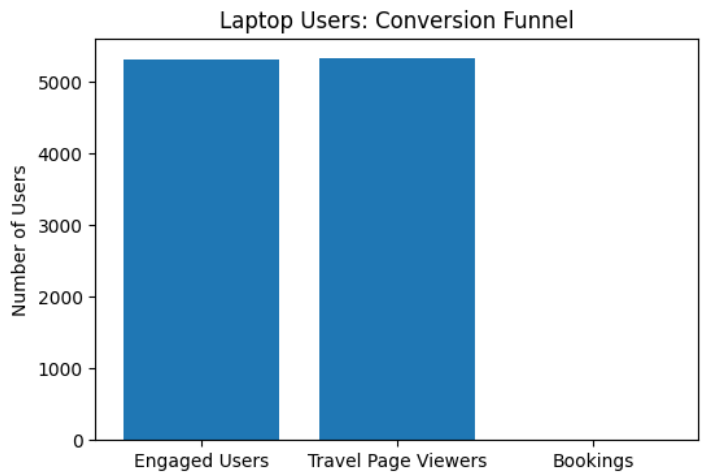


1. A large number of mobile users are engaged and view the travel page.
2. The number of users drops sharply at the booking stage.
3. This shows a high drop-off between page viewing and final booking.
4. Mobile users reach the funnel easily but fail to convert at the last step.
5. Improving the mobile booking and payment experience can increase conversions.

Funnel For Laptop Users

Observations:

Fig 15: Laptop-Conversion Funnel



1. A high number of laptop users are engaged and view the travel page.
2. There is a drop at the booking stage, but it is less severe than mobile users.
3. Laptop users move through the funnel more smoothly.
4. Higher screen size and ease of navigation likely support better conversion.
5. Laptop platforms show stronger booking potential compared to mobile.

Impact of Engagement on Ticket Purchase

Observations:

Fig 16: Impact of Engagement on Ticket Purchase

```
df_mobile.groupby('Taken_product')['Daily_Avg_mins_spend_on_traveling_page'] \
        .agg(['mean', 'median', 'skew'])
```

Taken_product	mean	median	skew
0	14.295570	12.0	6.645269
1	10.946557	9.0	6.618564

```
df_laptop.groupby('Taken_product')['Daily_Avg_mins_spend_on_traveling_page'] \
        .agg(['mean', 'median', 'skew'])
```

Taken_product	mean	median	skew
0	14.448112	13.0	0.880757
1	10.735165	9.0	1.397217

1. For mobile users, non-bookers spent more time on the travel page (14.3 mins) compared to bookers (10.9 mins), showing that higher time spent does not guarantee ticket purchase.
2. For laptop users, non-bookers also spent more time (14.4 mins) than bookers (10.7 mins), indicating similar behavior across devices.
3. Mobile users show higher variation in time spent (high skewness), suggesting more browsing without booking.
4. Overall, customers who book tickets tend to have shorter and more focused sessions on both devices.

Impact of Device Type on Ticket Purchase

Table 7: Impact of Device Type on Ticket Purchase

Taken_product	0	1
device_group		
Laptop	0.829364	0.170636
Mobile	0.845014	0.154986
Other	0.912752	0.087248

Observations:

1. Laptop users have a higher ticket purchase rate (17.1%) compared to mobile users (15.5%).
2. Mobile users show a slightly lower conversion, with 84.5% not purchasing tickets.
3. The “Other” device category has the lowest purchase rate (8.7%) and is not significant for analysis.
4. Overall, laptop users are more likely to purchase tickets than mobile users, making device type an important factor in predicting purchase behavior.

Data Pre-Processing

Data Preprocessing includes the data for modelling by:

1. **Checking for duplicates.**
2. **Checking for missing values.**
3. **Outlier detection and treatment**
4. **Feature engineering.**
5. **Data preparation for modeling.**

Checking for duplicates

Observation:

No duplicates are found in both the dataset.

Table 8: Checking Duplicate Values

```
df_mobile.duplicated().sum()
```

```
np.int64(0)
```

```
df_laptop.duplicated().sum()
```

```
np.int64(0)
```

Missing value treatment

Observations:

There are no missing values in both the dataset.

```
df_laptop.isnull().sum()
```

	0
UserID	0
Taken_product	0
Yearly_avg_view_on_travel_page	0
preferred_device	0
total_likes_on_outstation_checkin_given	0
yearly_avg_Outstation_checkins	0
member_in_family	0
preferred_location_type	0
Yearly_avg_comment_on_travel_page	0
total_likes_on_outofstation_checkin_received	0
week_since_last_outstation_checkin	0
following_company_page	0
montly_avg_comment_on_company_page	0
working_flag	0
travelling_network_rating	0
Adult_flag	0
Daily_Avg_mins_spend_on_traveling_page	0
device_group	0

dtype: int64

```
df_mobile.isnull().sum()
```

	0
UserID	0
Taken_product	0
Yearly_avg_view_on_travel_page	0
preferred_device	0
total_likes_on_outstation_checkin_given	0
yearly_avg_Outstation_checkins	0
member_in_family	0
preferred_location_type	0
Yearly_avg_comment_on_travel_page	0
total_likes_on_outofstation_checkin_received	0
week_since_last_outstation_checkin	0
following_company_page	0
montly_avg_comment_on_company_page	0
working_flag	0
travelling_network_rating	0
Adult_flag	0
Daily_Avg_mins_spend_on_traveling_page	0
device_group	0

dtype: int64

Table 9: Checking Missing Values

Outlier detection and treatment

For Mobile Users

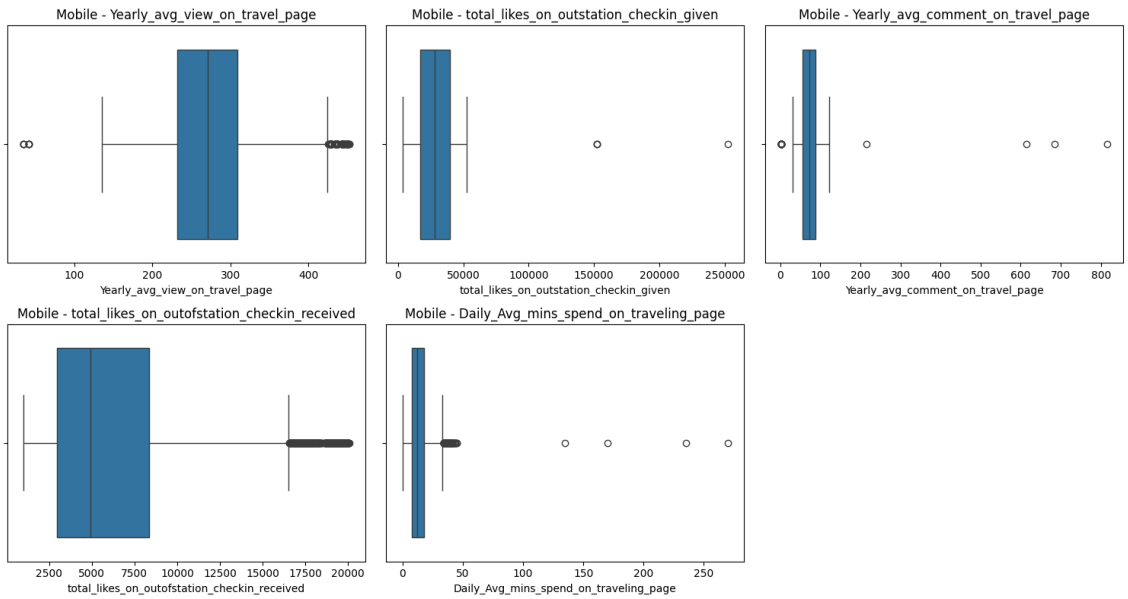


Fig 17: Outlier Detection For Mobile Users

Observations:

For mobile users, outliers are observed in engagement-related variables, indicating a small group of highly active users. These values represent valid customer behavior and were retained for modeling.

For Laptop Users

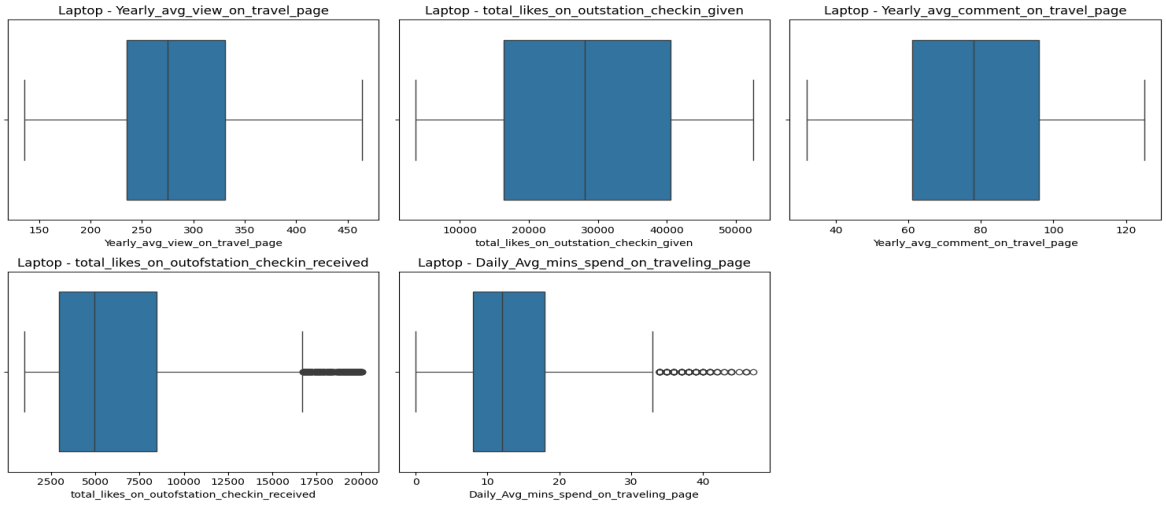


Fig 18: Outlier Detection For Laptop Users

Observations:

Laptop users show relatively wider spread in session duration and page views. The presence of extreme values reflects genuine high-intent behavior rather than noise.

IQR Count Summary

Note:

1. Outliers were not removed as they represent valid extreme customer behavior. Removing them could result in loss of important high-intent signals, especially for purchase prediction.
2. Discrete numerical variables such as Adult_flag, travelling_network_rating, yearly_avg_Outstation_checkins, and Taken_product were treated as categorical. Hence, boxplots and correlation-based outlier detection were not applied to them.

Fig 19: IQR Count Summary

```python
def outlier_count_iqr(df, cols):
    for col in cols:
        Q1 = df[col].quantile(0.25)
        Q3 = df[col].quantile(0.75)
        IQR = Q3 - Q1
        lower = Q1 - 1.5 * IQR
        upper = Q3 + 1.5 * IQR
        count = df[(df[col] < lower) | (df[col] > upper)].shape[0]
        print(f"{col}: {count} outliers")

print("Mobile Outliers")
outlier_count_iqr(df_mobile, continuous_cols)

print("\nLaptop Outliers")
outlier_count_iqr(df_laptop, continuous_cols)
```

```
Mobile Outliers
Yearly_avg_view_on_travel_page: 132 outliers
total_likes_on_outstation_checkin_given: 4 outliers
Yearly_avg_comment_on_travel_page: 40 outliers
total_likes_on_outofstation_checkin_received: 489 outliers
Daily_Avg_mins_spend_on_traveling_page: 180 outliers

Laptop Outliers
Yearly_avg_view_on_travel_page: 0 outliers
total_likes_on_outstation_checkin_given: 0 outliers
Yearly_avg_comment_on_travel_page: 0 outliers
total_likes_on_outofstation_checkin_received: 396 outliers
Daily_Avg_mins_spend_on_traveling_page: 163 outliers
```

Feature Engineering

To improve prediction of customer ticket purchase behavior, key variables were transformed and engineered into business-meaningful indicators.

Feature Creation

- A **Social Engagement Score** was created by combining likes and check-in interactions to capture overall social activity on the platform.
- An **Engagement Intensity** metric aggregates page views, comments, and time spent, reflecting depth of interest in travel content.
- A **Recent Traveler Flag** identifies users who traveled in the last 4 weeks, signaling strong current travel intent.

- Binary flags for working status were created for clear interpretation and modeling.

Data Cleaning & Transformation

- Inconsistent values in family size text entries were standardized for accuracy.
- Redundant and identifier columns were removed to reduce noise and prevent data leakage.

Encoding Categorical Variables

- Categorical fields such as preferred location type were one-hot encoded to make them usable for machine learning models.

Overall, these steps simplified complex user behavior into actionable features, improving model accuracy and business interpretability.

Preparing Data For Modeling

- Splitting the data into dependent(y) and independent variable(X) for both mobile and laptop dataframe.
    - Taken_product as a dependent variable(y).
    - All the remaining columns except Taken_product as independent variable(X).
- **add_constants()** used to add constant terms to the dataset.
- Splitting the data in a 80:20 ratio for train and test data using **train_test_split()**.
- Numeric columns were scaled so all values are in the same range.
- Scaling was done using only training data to avoid data leakage.

Baseline Model Building

Logistic Regression

For Mobile

```
LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=1000)
```

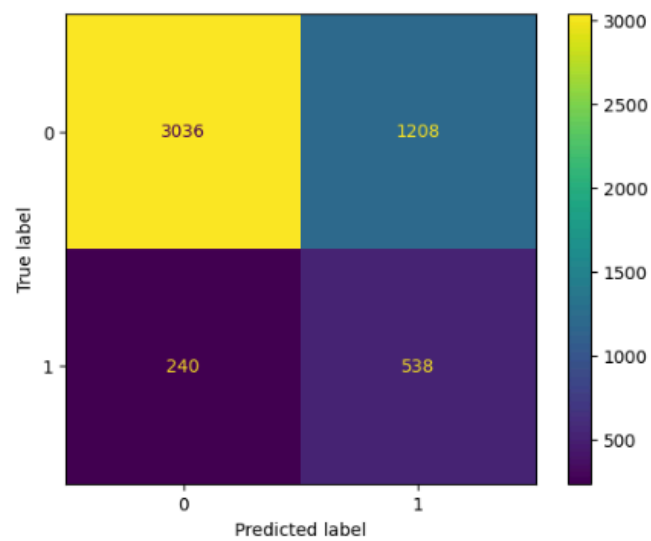
Fig 20:Mobile-Logistic Regression

Confusion Matrix and Performance Check of Training data - For Mobile

Fig 21: Mobile-Confusion Matrix and Performance Check

	precision	recall	f1-score	support
0	0.93	0.72	0.81	4244
1	0.31	0.69	0.43	778
accuracy			0.71	5022
macro avg	0.62	0.70	0.62	5022
weighted avg	0.83	0.71	0.75	5022

```
np.float64(0.766802793116064)
```



Observations:

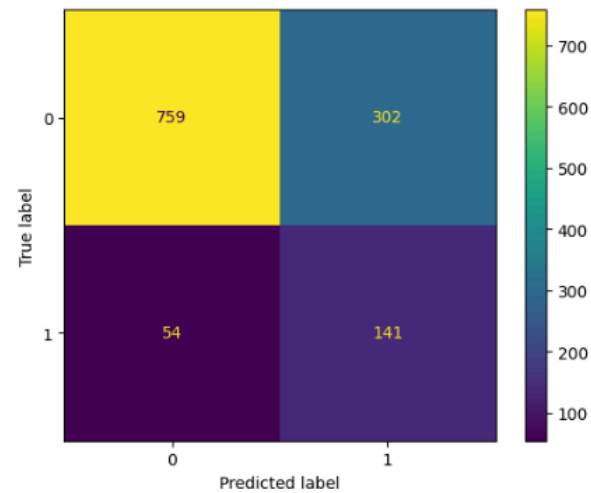
1. The model captures most mobile ticket buyers (Recall = 0.69), so fewer real buyers are missed.
2. Precision is low (0.31), meaning many users predicted as buyers do not actually purchase.
3. Confusion matrix shows 538 real buyers identified, but 1,208 non-buyers wrongly flagged as buyers.
4. F1-score = 0.43 indicates moderate balance between catching buyers and false alarms.
5. ROC-AUC ≈ 0.77, showing the model can fairly distinguish buyers vs non-buyers on mobile.

Confusion Matrix and Performance Check of Testing data - For Mobile

Fig 22: Mobile-Confusion Matrix and Performance Check

	precision	recall	f1-score	support
0	0.93	0.72	0.81	1061
1	0.32	0.72	0.44	195
accuracy			0.72	1256
macro avg	0.63	0.72	0.63	1256
weighted avg	0.84	0.72	0.75	1256

np.float64(0.780434519925566)



Observations:

1. The model identifies most actual mobile ticket buyers (Recall = 0.72), so few buyers are missed.
2. Many users predicted that buyers do not actually buy (Precision = 0.32).
3. Confusion matrix shows 141 real buyers found, but 302 non-buyers wrongly flagged as buyers.
4. ROC-AUC ≈ 0.78, meaning the model is fairly good at separating buyers and non-buyers.
5. Mobile is useful for promotions and reminders, but predictions should not be treated as guaranteed purchases.

For Laptop

```
LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=2000)
```

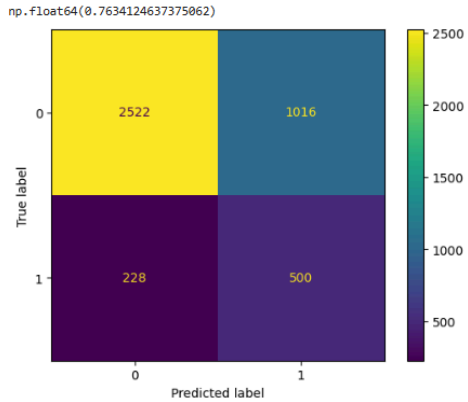
Fig 23:Mobile-Logistic Regression

Confusion Matrix and Performance Check of Training data - For Laptop

Fig 24: Laptop-Confusion Matrix and Performance Check

	precision	recall	f1-score	support
0	0.92	0.71	0.80	3538
1	0.33	0.69	0.45	728
accuracy			0.71	4266
macro avg	0.62	0.70	0.62	4266
weighted avg	0.82	0.71	0.74	4266

np.float64(0.7634124637375062)



Observations:

1. The model identifies most laptop ticket buyers (Recall = 0.69), so many real buyers are captured.
2. Precision is low (0.33), meaning several users predicted as buyers do not actually purchase.
3. Confusion matrix shows 500 real buyers correctly identified, but 1,016

non-buyers wrongly predicted as buyers.

4. F1-score = 0.45 indicates moderate performance in balancing buyer detection and false predictions.
5. ROC-AUC ≈ 0.76, showing the model can reasonably distinguish buyers from non-buyers on a laptop.

Confusion Matrix and Performance Check of Testing data - For Laptop

Observations:

1. The model identifies most real laptop ticket buyers (Recall = 0.70), so fewer buyers are missed.
2. Precision is low (0.34), meaning many users predicted as buyers do not actually purchase.
3. Confusion matrix shows 127 real buyers correctly identified, but 242 non-buyers wrongly flagged as buyers.
4. F1-score = 0.46 indicates moderate and balanced performance.
5. ROC-AUC ≈ 0.77, showing the model can fairly distinguish buyers from non-buyers on laptops.

	precision	recall	f1-score	support
0	0.92	0.73	0.81	885
1	0.34	0.70	0.46	182
accuracy			0.72	1067
macro avg	0.63	0.71	0.64	1067
weighted avg	0.82	0.72	0.75	1067

```
np.float64(0.7710188116967778)
```

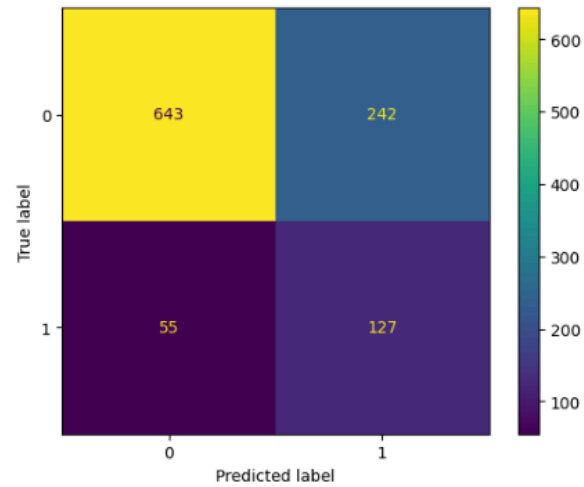


Fig 25: Laptop-Confusion Matrix and Performance Check

Advance Model Building with Hyperparameter Tuning

Decision Tree Model

For Mobile

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=1)
```

Fig 26:Mobile-Decision Tree Model

Confusion Matrix and Performance Check of Testing data - For Mobile

Observations:

1. The model performs very well overall (Accuracy ≈ 0.93), showing strong prediction quality.
2. High precision (≈ 0.79) means most users predicted that buyers would actually buy tickets.
3. Recall ≈ 0.73 indicates the model captures most real mobile buyers, with few missed.
4. F1-score ≈ 0.76 shows a good balance between precision and recall.
5. ROC-AUC ≈ 0.85, meaning the Decision Tree clearly separates buyers from non-buyers.

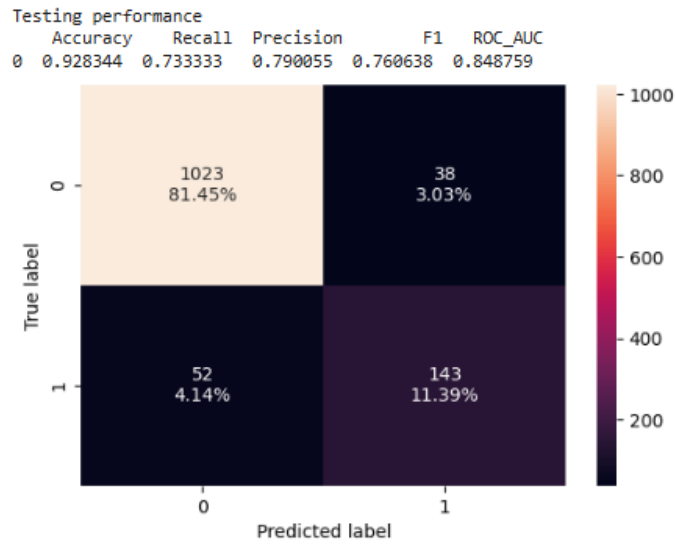


Fig 27:Mobile-Confusion Matrix and Performance Check

Confusion Matrix and Performance Check of Training data - For Mobile

Observations:

1. The model shows perfect performance on training data (Accuracy, Precision, Recall, F1, ROC-AUC = 1.0).
2. The confusion matrix shows no misclassifications – all buyers and non-buyers are predicted correctly.
3. This indicates overfitting, not real-world perfection.
4. Such perfect scores mean the model has memorized training data.
5. Predictions on new mobile users may be less reliable than training results suggest.

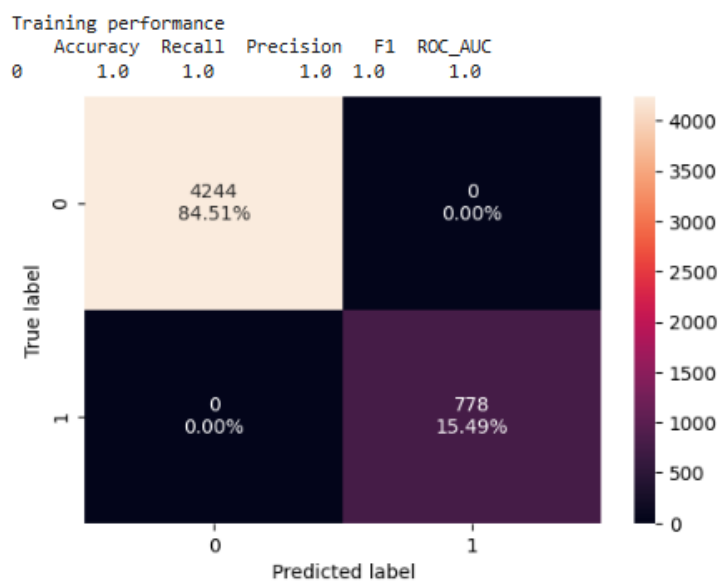


Fig 28:Mobile-Confusion Matrix and Performance Check

Decision Tree with Hyperparameter Tuning - For Mobile

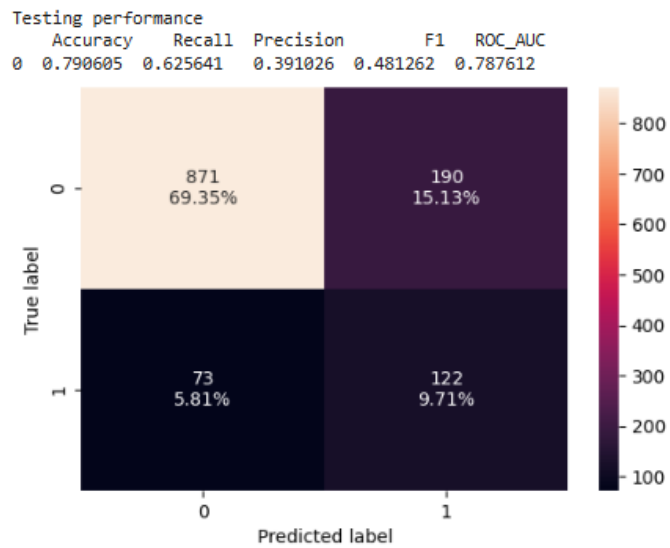
```
DecisionTreeClassifier
DecisionTreeClassifier(class_weight='balanced', max_depth=np.int64(8),
                       max_leaf_nodes=np.int64(16),
                       min_impurity_decrease=0.0001, min_samples_split=50,
                       random_state=1)
```

Fig 29: Mobile- Decision Tree with Hyperparameter Tuning

Confusion Matrix and Performance Check After Tuning - For Mobile

Testing Data

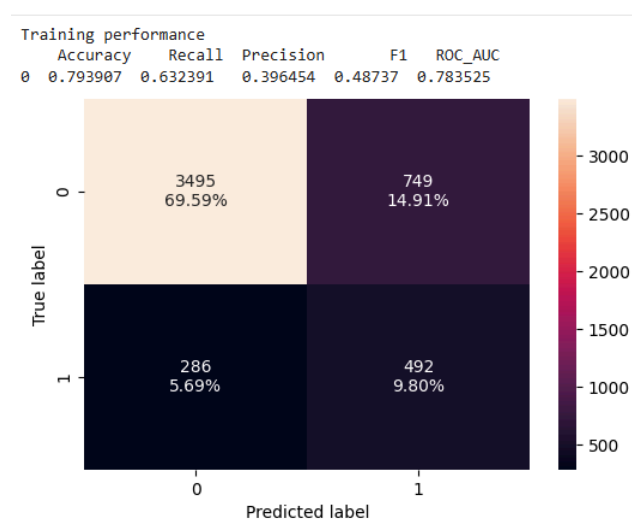
Fig 30:Mobile:Confusion Matrix and Performance Check



1. ROC-AUC ≈ 0.79, showing the model can fairly distinguish buyers from non-buyers.
2. Recall ≈ 0.63, meaning the model catches most real mobile buyers, but some are still missed.
3. Precision ≈ 0.39, so several users predicted as buyers do not actually buy.
4. Confusion matrix shows 122 real buyers correctly identified, while 190 non-buyers are wrongly flagged.
5. F1-score ≈ 0.48, indicating moderate and balanced performance.

Training Data

Fig 31:Mobile:Confusion Matrix and Performance Check



1. ROC-AUC ≈ 0.78, consistent with test performance (good sign).
2. Recall ≈ 0.63, showing the model learns buyer patterns reasonably well.
3. Precision ≈ 0.40, meaning false buyer predictions still exist.
4. Confusion matrix shows 492 real buyers identified, but 749 non-buyers wrongly predicted as buyers.
5. F1-score ≈ 0.49, indicating stable but moderate performance.

For Laptop

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=1)
```

Fig 32:Laptop-Decision Tree Model

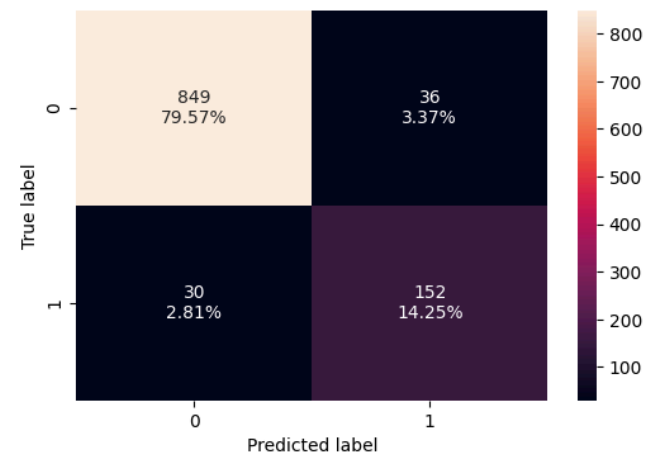
Confusion Matrix and Performance Check of Testing data - For Laptop

Observations:

1. Very strong performance with Accuracy ≈ 0.94.
2. High Precision ≈ 0.81, meaning most users predicted as buyers actually buy tickets.
3. High Recall ≈ 0.84, so the model captures most real laptop buyers.
4. Confusion matrix shows 152 real buyers correctly identified, with very few buyers missed (30).
5. ROC-AUC ≈ 0.90, indicating excellent separation between buyers and non-buyers.

Fig 33:Laptop-Confusion Matrix and Performance Check

```
Testing performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.938144  0.835165   0.808511  0.821622  0.897243
```



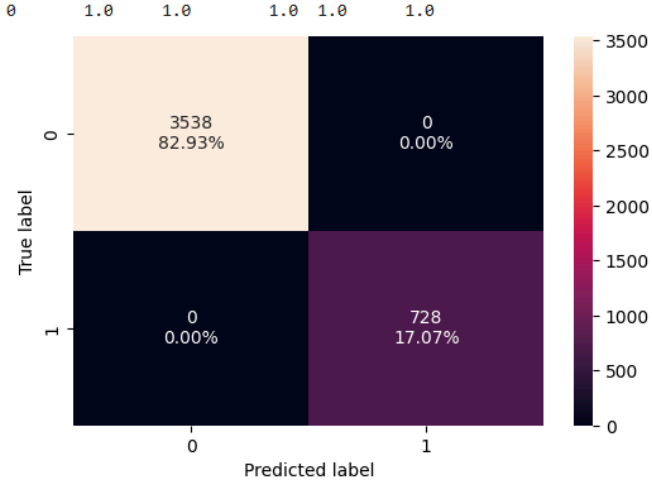
Confusion Matrix and Performance Check of Training data - For Laptop

Observations:

1. Perfect scores (Accuracy, Precision, Recall, F1, ROC-AUC = 1.0).
2. The confusion matrix shows zero misclassifications.
3. This clearly indicates overfitting.
4. The model has memorized training data, not learned general patterns.
5. Train results should not be trusted alone for business decisions.

Fig 34:Laptop-Confusion Matrix and Performance Check

```
Training performance
    Accuracy  Recall  Precision   F1  ROC_AUC
0       1.0     1.0        1.0  1.0      1.0
```



Decision Tree with Hyperparameter Tuning - For Laptop

```
DecisionTreeClassifier
DecisionTreeClassifier(class_weight='balanced', max_depth=np.int64(8),
                       max_leaf_nodes=np.int64(18),
                       min_impurity_decrease=0.0001, min_samples_split=50,
                       random_state=1)
```

Fig 35: Laptop- Decision Tree with Hyperparameter Tuning

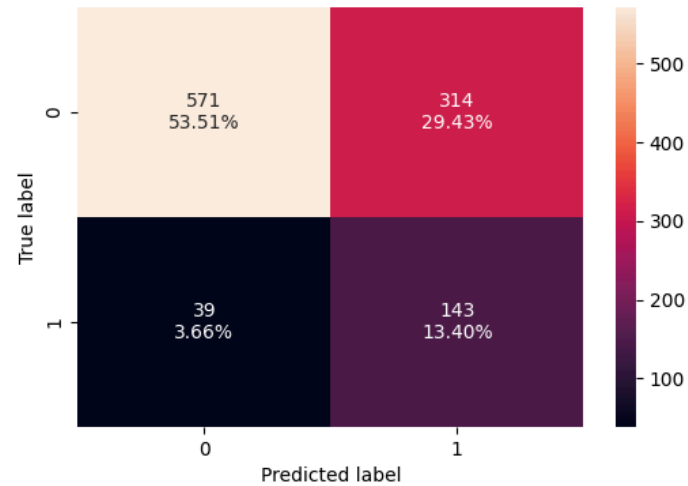
Confusion Matrix and Performance Check After Tuning - For Laptop

Testing Data

1. Moderate overall performance (Accuracy ≈ 0.67).
2. High Recall ≈ 0.79, so the model catches most real laptop buyers.
3. Low Precision ≈ 0.31, meaning many predicted buyers do not actually buy.
4. Confusion matrix shows 143 real buyers identified, but 314 non-buyers wrongly predicted as buyers.
5. ROC-AUC ≈ 0.76, indicating fair buyer vs non-buyer separation.

Fig 36:Laptop-Confusion Matrix and Performance Check

```
Testing performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.669166  0.785714    0.31291  0.447574  0.758872
```

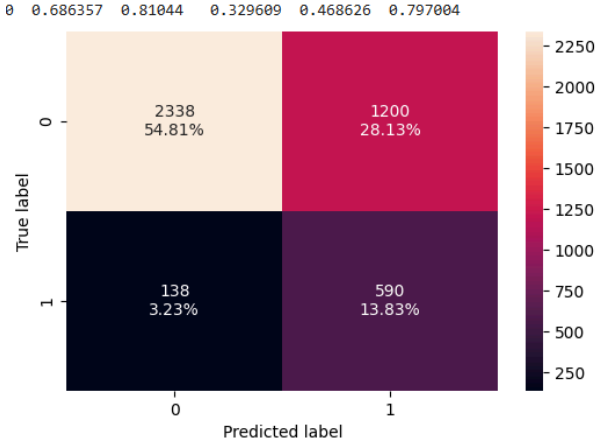


Training Data

1. Similar performance to test data (Accuracy ≈ 0.69), showing reduced overfitting.
2. Recall ≈ 0.81, consistently capturing laptop buyers.
3. Precision ≈ 0.33, false buyer predictions remain high.
4. ROC-AUC ≈ 0.80, slightly better discrimination on training data.
5. F1-score ≈ 0.47, showing moderate, stable balance.

Fig 37:Laptop-Confusion Matrix and Performance Check

```
Training performance
    Accuracy   Recall  Precision        F1   ROC_AUC
0  0.686357  0.81044   0.329609  0.468626  0.797004
```



Bagging Classifier

For Mobile

```
BaggingClassifier
BaggingClassifier(random_state=1)
```

Fig 38:Mobile-Bagging Classifier Model

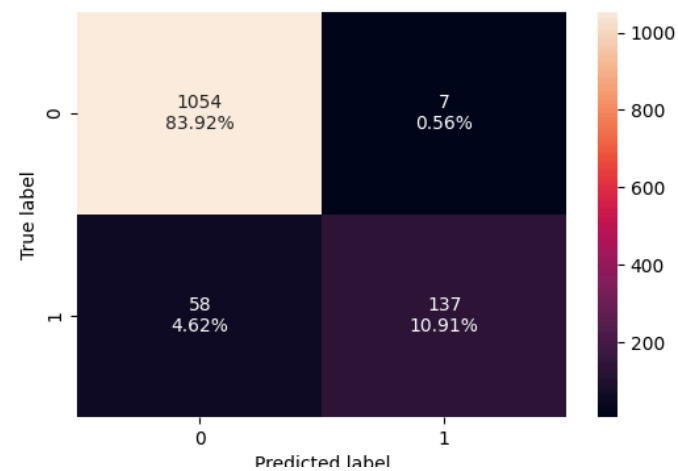
Confusion Matrix and Performance Check of Testing data - For Mobile

Observations:

1. Excellent overall performance (Accuracy ≈ 0.95).
2. Very high Precision ≈ 0.95, meaning almost all users predicted as buyers actually buy tickets.
3. Good Recall ≈ 0.70, so most real buyers are captured, with some still missed.
4. Confusion matrix shows 137 real buyers correctly identified and very few false buyer predictions (only 7).
5. ROC-AUC ≈ 0.96, indicating outstanding separation between buyers and non-buyers.

Fig 39:Mobile-Confusion Matrix and Performance Check

```
Testing performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.948248  0.702564   0.951389  0.80826  0.963423
```



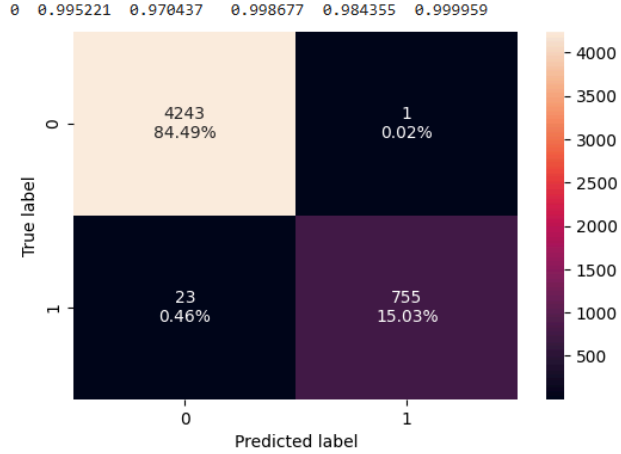
Confusion Matrix and Performance Check of Training data - For Mobile

Observations:

1. Near-perfect performance (Accuracy ≈ 0.995, F1 ≈ 0.98).
2. The confusion matrix shows almost no misclassifications.
3. ROC-AUC ≈ 1.00, showing extremely strong learning.
4. Indicates very strong fit, but test performance confirms it generalizes well.
5. Train results support the robustness of the model.

Fig 40:Mobile-Confusion Matrix and Performance Check

```
Training performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.995221  0.970437   0.998677  0.984355  0.999959
```



Bagging Classifier With Hyperparameter Tuning - For Mobile

```
BaggingClassifier
BaggingClassifier(max_features=0.7, max_samples=0.7, n_estimators=60,
                  random_state=1)
```

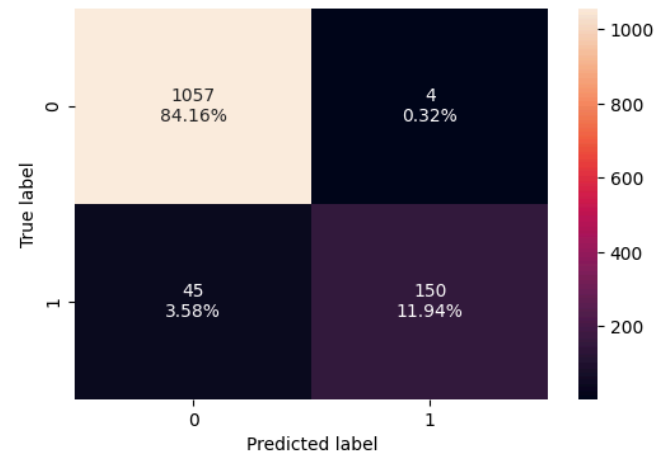
Fig 41:Mobile-Tuned Bagging Classifier Model

Confusion Matrix and Performance Check After Tuning - For Mobile

Testing Data

Fig 42:Mobile-Confusion Matrix and Performance Check

```
Testing performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.960987  0.769231   0.974026  0.859599  0.988407
```

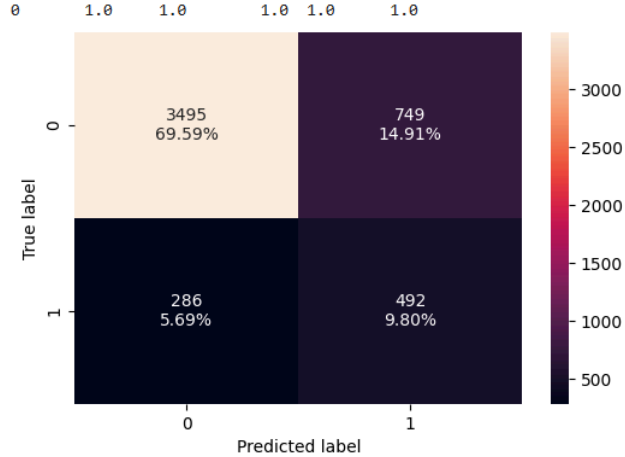


1. Excellent overall performance (Accuracy ≈ 0.96).
2. Very high Precision ≈ 0.97, meaning almost all users predicted as buyers actually purchase tickets.
3. Good Recall ≈ 0.77, so most real mobile buyers are captured, with a few missed.
4. Confusion matrix shows 150 real buyers correctly identified and only 4 non-buyers wrongly predicted as buyers.
5. ROC-AUC ≈ 0.99, indicating outstanding ability to separate buyers from non-buyers.

Training Data

Fig 42:Mobile-Confusion Matrix and Performance Check

```
Training performance
    Accuracy  Recall  Precision   F1  ROC_AUC
0       1.0     1.0        1.0  1.0      1.0
```



1. Perfect training performance (Accuracy, Precision, Recall, F1, ROC-AUC = 1.0).
2. Confusion matrix shows no misclassifications on training data.
3. Indicates very strong fitting.
4. Strong test results confirm the model generalizes well.
5. Overall, this is a robust and stable model.

For Laptop

```
BaggingClassifier
BaggingClassifier(random_state=1)
```

Fig 43:Laptop-Bagging Classifier Model

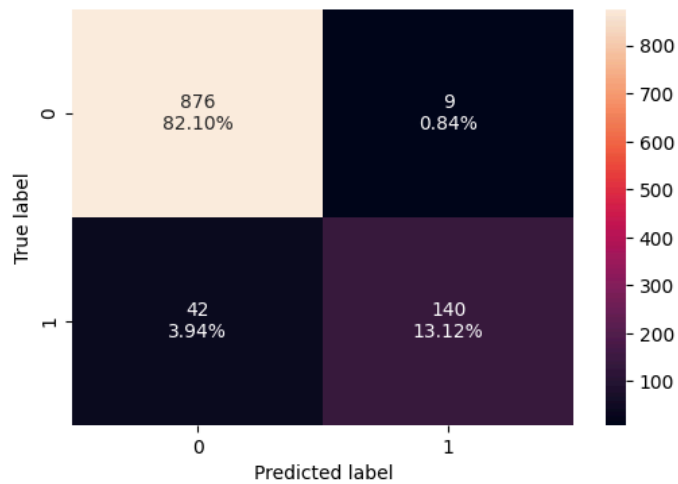
Confusion Matrix and Performance Check of Testing data - For Laptop

Observations:

1. Excellent overall performance with Accuracy ≈ 0.94.
2. High Precision ≈ 0.81, so most users predicted that buyers actually purchase tickets.
3. Strong Recall ≈ 0.84, meaning the model captures most real laptop buyers.
4. Confusion matrix shows 140 real buyers correctly identified, with very few false buyer predictions (9).
5. ROC-AUC ≈ 0.90, indicating very good separation between buyers and non-buyers.

Fig 44:Laptop-Confusion Matrix and Performance Check

```
Testing performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.938144  0.835165   0.808511  0.821622  0.897243
```



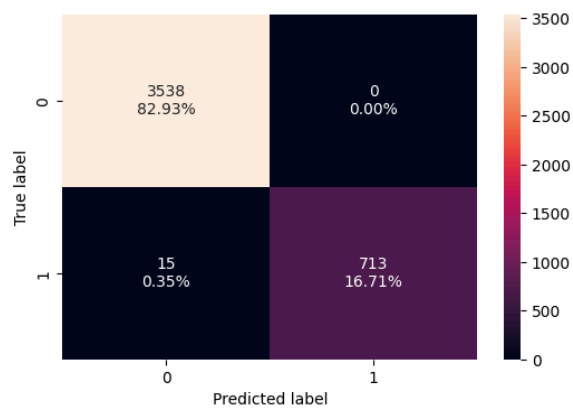
Confusion Matrix and Performance Check of Training data - For Laptop

Observations:

1. Perfect training performance (Accuracy, Precision, Recall, F1, ROC-AUC = 1.0).
2. The confusion matrix shows no misclassifications.
3. Indicates very strong fitting, but
4. Strong test performance confirms the model generalizes well.
5. Train results support the stability and robustness of the model.

Fig 45:Laptop-Confusion Matrix and Performance Check

```
Training performance
    Accuracy  Recall  Precision   F1  ROC_AUC
0        1.0     1.0        1.0  1.0      1.0
```



Bagging Classifier with Hyperparameter Tuning - For Laptop

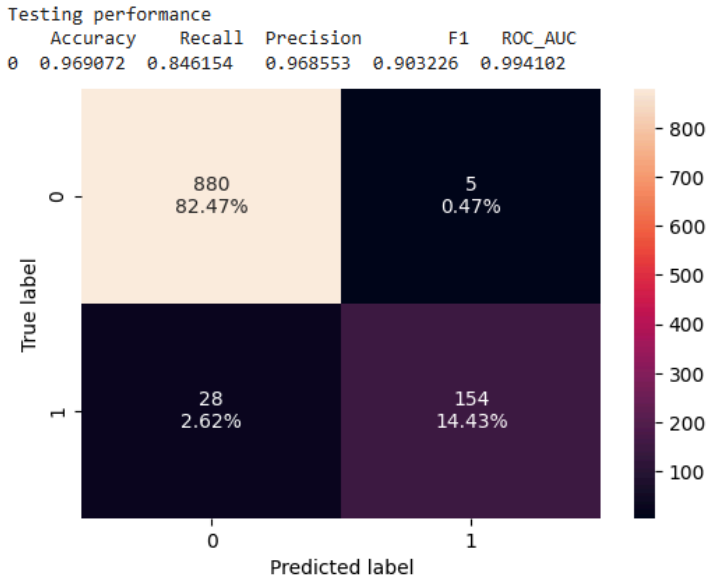
```
BaggingClassifier
BaggingClassifier(max_features=0.7, max_samples=0.7, n_estimators=60,
                  random_state=1)
```

Fig 46:Laptop-Tuned Bagging Classifier Model

Confusion Matrix and Performance Check After Tuning - For Laptop

Testing Data

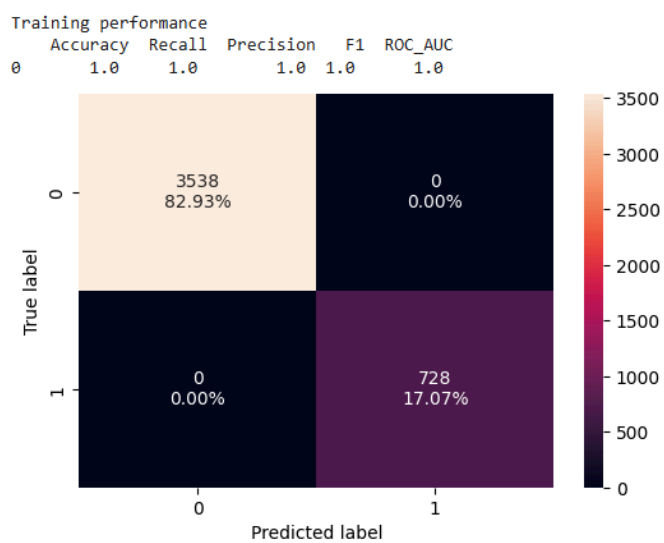
Fig 47:Laptop-Confusion Matrix and Performance Check



1. Outstanding overall performance (Accuracy ≈ 0.97).
2. Very high Precision ≈ 0.97, meaning almost every user predicted as a buyer actually buys a ticket.
3. High Recall ≈ 0.85, so most real laptop buyers are successfully captured.
4. Confusion matrix shows 154 real buyers correctly identified, with only 5 non-buyers wrongly predicted as buyers.
5. ROC-AUC ≈ 0.99, indicating excellent separation between buyers and non-buyers.

Training Data

Fig 48:Laptop-Confusion Matrix and Performance Check



1. Perfect training performance (Accuracy, Precision, Recall, F1, ROC-AUC = 1.0).
2. The confusion matrix shows zero misclassifications.
3. Indicates very strong fitting.
4. Strong test performance confirms the model generalizes very well.
5. Training results support the robustness and stability of the model.

Random Forest Classifier

For Mobile

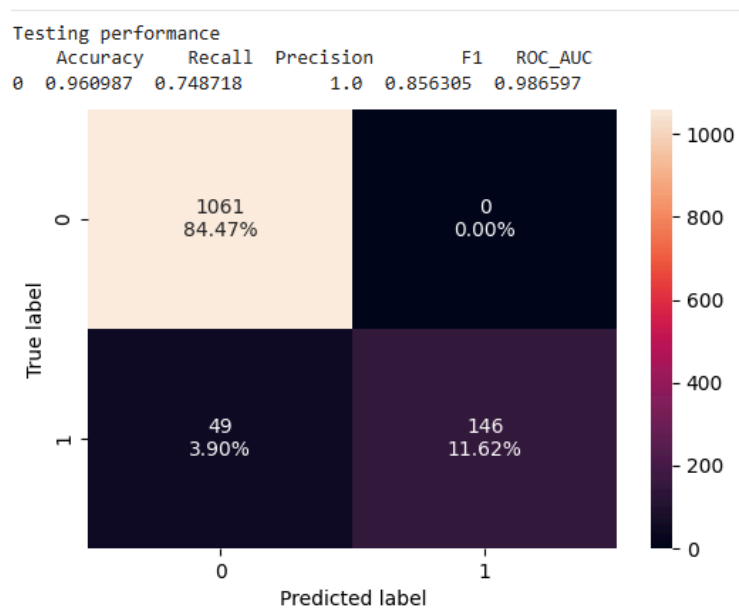
```
RandomForestClassifier
RandomForestClassifier(random_state=1)
```

Fig 49:Mobile-Random Forest Classifier Model

Confusion Matrix and Performance Check of Testing data - For Mobile

Observations:

Fig 50:Mobile-Confusion Matrix and Performance Check

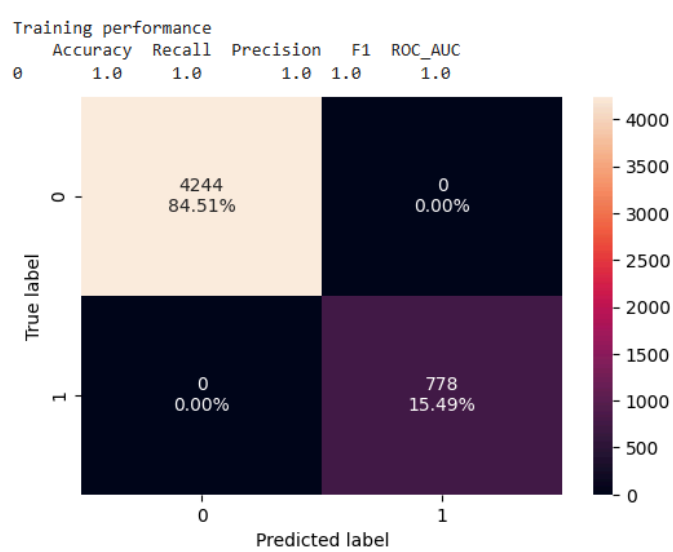


1. Very strong overall performance (Accuracy ≈ 0.96).
2. Perfect Precision = 1.0, meaning every user predicted as a buyer actually bought a ticket (no false positives).
3. Recall ≈ 0.75, so most real mobile buyers are captured, though a few are missed.
4. Confusion matrix shows 146 real buyers correctly identified and 0 non-buyers wrongly predicted as buyers.
5. ROC-AUC ≈ 0.99, indicating excellent separation between buyers and non-buyers.

Confusion Matrix and Performance Check of Training data - For Mobile

Observations:

Fig 51:Mobile-Confusion Matrix and Performance Check



1. Perfect training performance (Accuracy, Precision, Recall, F1, ROC-AUC = 1.0).
2. The confusion matrix shows no misclassifications.
3. Indicates strong fitting.
4. Excellent test performance confirms the model generalizes very well.
5. Training results support the robustness of the model.

Random Forest With Hyperparameter Tuning - For Mobile

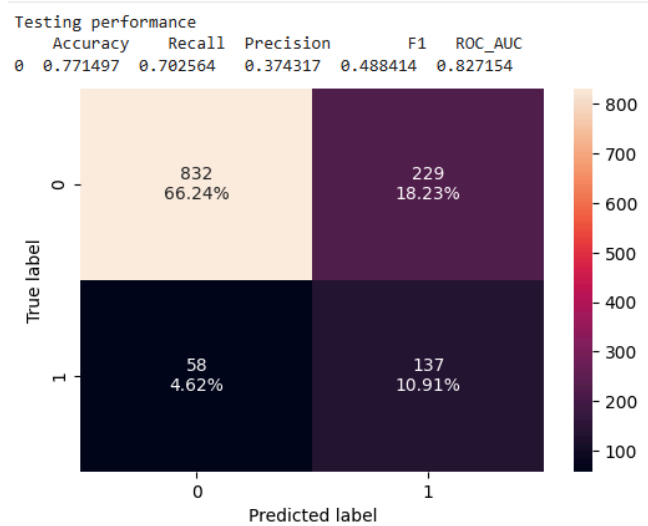
```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', max_depth=np.int64(4),
                       max_samples=0.7, min_samples_split=50,
                       n_estimators=np.int64(80), random_state=1)
```

Fig 52:Mobile-Tuned Random Forest Model

Confusion Matrix and Performance Check After Tuning - For Mobile

Testing Data

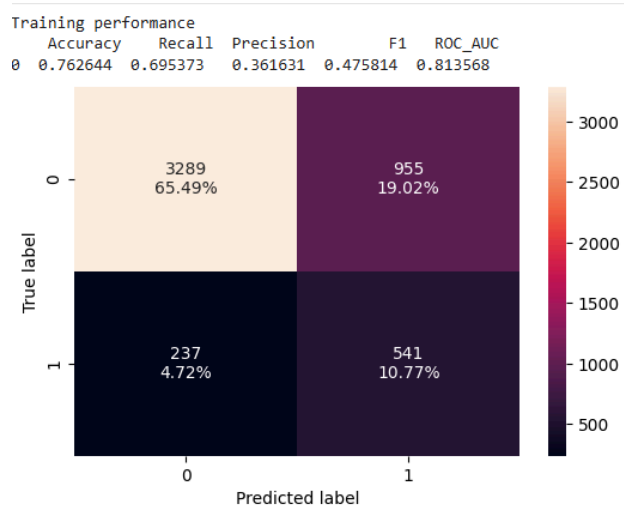
Fig 53:Mobile-Confusion Matrix and Performance Check



1. Moderate overall accuracy (~77%), showing reasonable prediction performance.
2. Precision ≈ 0.37 → when the model predicts a mobile user will buy a ticket, it is correct only about 1 out of 3 times (many false positives).
3. Recall ≈ 0.70 → the model is able to identify most actual buyers, which is good for capturing potential customers.
4. Confusion matrix shows 137 true buyers correctly predicted, but 229 non-buyers wrongly predicted as buyers.
5. ROC-AUC ≈ 0.83, indicating good class separation, though weaker than the untuned model.

Training Data

Fig 54:Mobile-Confusion Matrix and Performance Check



1. Accuracy ≈ 76%, close to test performance → no severe overfitting.
2. Precision ≈ 0.36, still relatively low, confirming false positives remains an issue.
3. Recall ≈ 0.70, consistently captures a majority of real buyers.
4. Confusion matrix shows many correct buyer predictions, but also a noticeable number of false alerts.
5. ROC-AUC ≈ 0.81, showing stable but moderate discrimination.

For Laptop

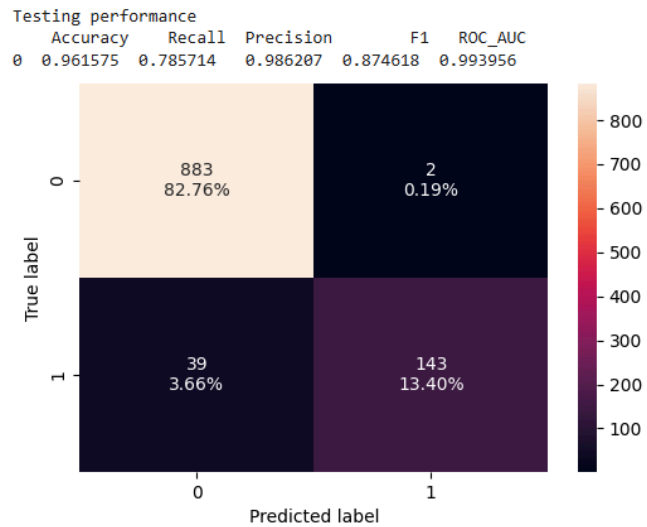
```
RandomForestClassifier
RandomForestClassifier(random_state=1)
```

Fig 55:Laptop-Random Forest Classifier Model

Confusion Matrix and Performance Check of Testing data - For Laptop

Observations:

Fig 56:Laptop-Confusion Matrix and Performance Check

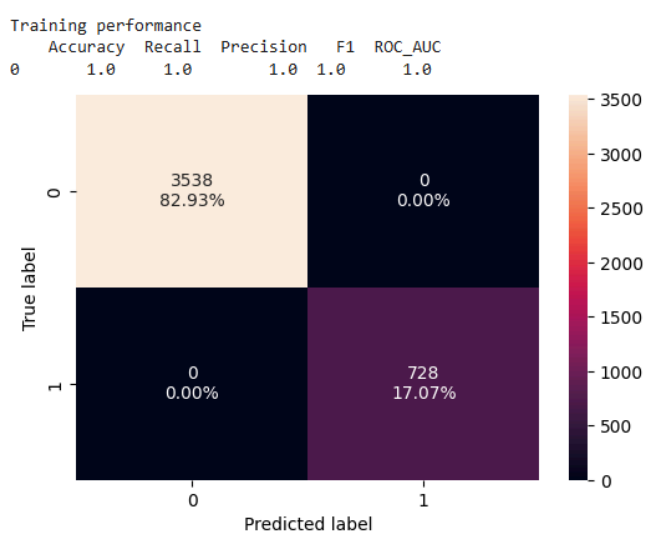


1. Very high accuracy (~96%), showing strong overall prediction performance.
2. Precision ≈ 0.99 → when the model predicts a laptop user will buy a ticket, it is almost always correct.
3. Recall ≈ 0.79 → the model correctly identifies most actual buyers, with very few missed.
4. Confusion matrix shows 143 true buyers correctly predicted and only 2 false positives, indicating excellent reliability.
5. ROC-AUC ≈ 0.99, showing excellent class separation between buyers and non-buyers.

Confusion Matrix and Performance Check of Training data - For Laptop

Observations:

Fig 57:Laptop-Confusion Matrix and Performance Check



1. Perfect accuracy, precision, recall, F1, and ROC-AUC (all = 1.0).
2. Confusion matrix shows zero misclassifications on training data.
3. This indicates possible overfitting, but test performance remains very strong.

Random Forest With Hyperparameter Tuning - For Laptop

```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', max_depth=np.int64(4),
                       max_samples=1.0, min_samples_split=50,
                       n_estimators=np.int64(80), random_state=1)
```

Fig 58:Laptop-Tuned Random Forest Model

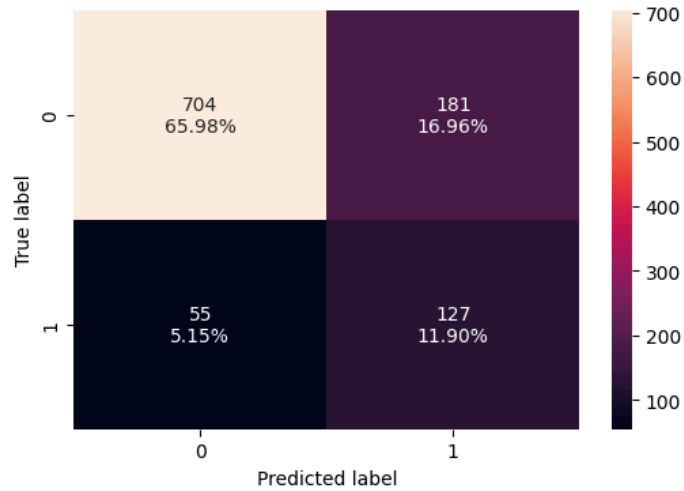
Confusion Matrix and Performance Check After Tuning - For Laptop

Testing Data

1. The model correctly identifies 127 buyers, which explains the high recall (69.8%).
2. However, 181 non-buyers are wrongly predicted as buyers, causing low precision (41.2%).
3. The moderate F1-score (0.52) shows imbalance between catching buyers and false alarms.
4. ROC-AUC (0.81) indicates good separation between buyers and non-buyers despite false positives

Fig 59:Laptop-Confusion Matrix and Performance Check

```
Testing performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.778819  0.697802   0.412338  0.518367   0.81472
```

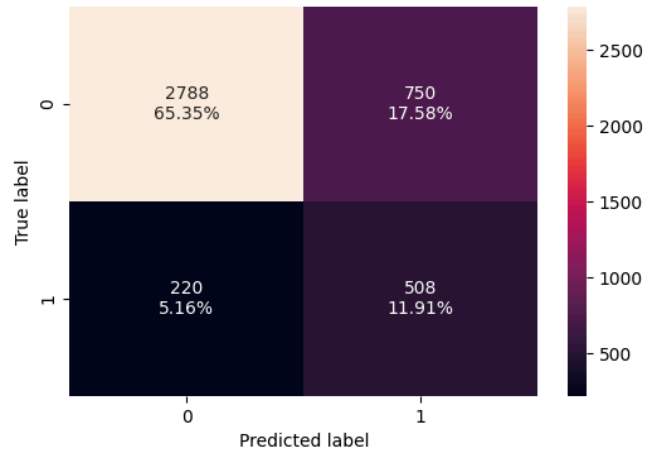


Training Data

1. The model correctly predicts 508
2. buyers, maintaining strong recall (69.8%).
3. A high number of false positives (750) reduces precision to 40.4%.
4. F1-score (0.51) confirms moderate overall buyer prediction quality.
5. ROC-AUC (0.82) shows the model still distinguishes buyers well overall.

Fig 60:Laptop-Confusion Matrix and Performance Check

```
Training performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.772621  0.697802   0.403816  0.511581  0.823464
```



AdaBoost Classifier

For Mobile

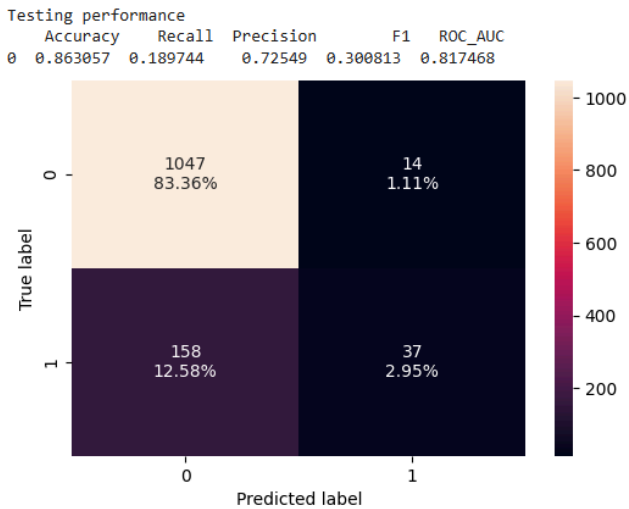
```
AdaBoostClassifier
AdaBoostClassifier(random_state=1)
```

Fig 61:Mobile-AdaBoost Classifier Model

Confusion Matrix and Performance Check of Testing data - For Mobile

Observations:

Fig 62:Mobile-Confusion Matrix and Performance Check

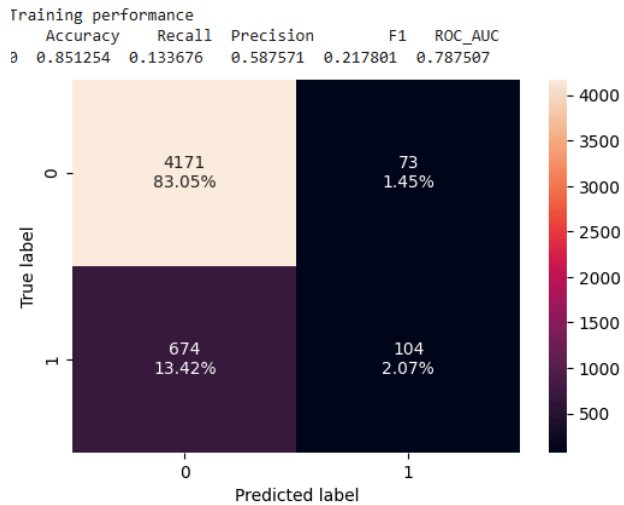


1. 1047 non-buyers are correctly predicted, which gives a high accuracy of 86 percent.
2. Only 37 buyers are correctly identified, while 158 buyers are missed, leading to a low recall of 19 percent.
3. Only 14 non-buyers are wrongly predicted as buyers, so precision is high at 73 percent.
4. Because many buyers are missed, the F1 score remains low at 0.30.
5. ROC score of 0.82 indicates good separation between buyers and non-buyers, but buyer prediction is conservative.

Confusion Matrix and Performance Check of Training data - For Mobile

Observations:

Fig 63:Mobile-Confusion Matrix and Performance Check



1. 4171 non-buyers are correctly predicted as non-buyers, which leads to a high accuracy of 85 percent.
2. 674 actual buyers are wrongly predicted as non-buyers, so the model misses most buyers, giving a very low recall of 13 percent.
3. Because the model predicts very few buyers incorrectly, precision is moderate at 59 percent, meaning predicted buyers are often correct.

4. The large gap between buyers found and buyers missed results in a low F1 score of 0.22.
5. ROC score of 0.79 shows the model can separate buyers and non-buyers reasonably well, but it avoids predicting buyers.

AdaBoost With Hyperparameter Tuning - For Mobile

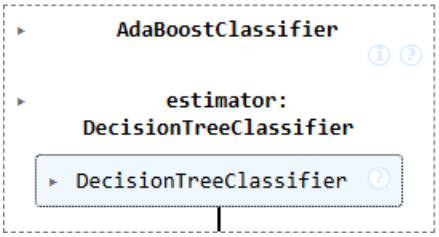
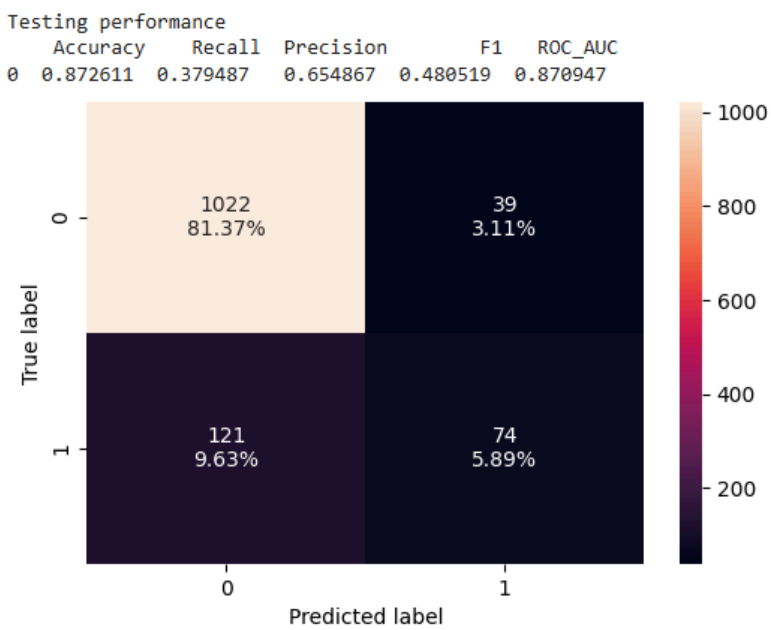


Fig 64:Mobile-Tuned AdaBoost Model

Confusion Matrix and Performance Check After Tuning - For Mobile

Testing Data

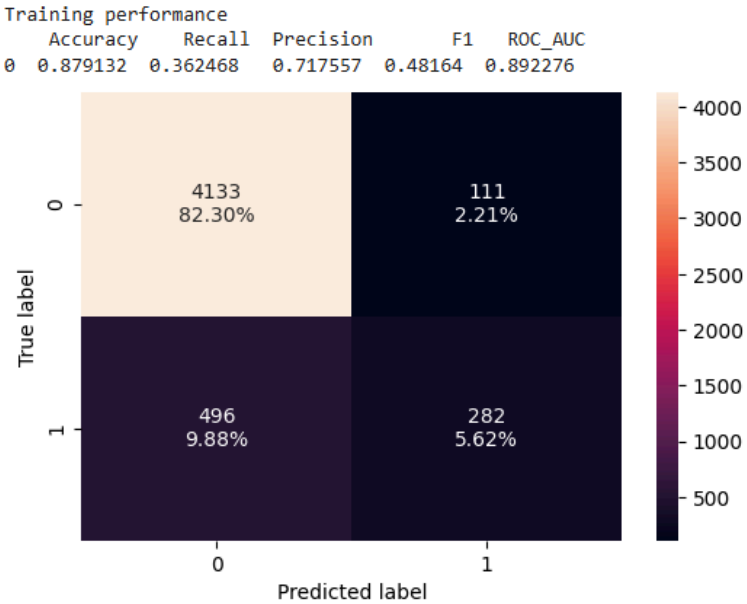
Fig 65:Mobile-Confusion Matrix and Performance Check



1. Confusion matrix shows 1022 non-buyers correctly predicted, 39 non-buyers wrongly predicted, 74 buyers correctly predicted, and 121 buyers wrongly predicted.
2. Accuracy of the model is 87.26 percent, showing stable performance on new data.
3. Recall is 37.95 percent, so most buyers are still not identified.
4. Precision is 65.49 percent, showing reasonable correctness of buyer predictions.
5. F1 score is 0.48 and ROC score is 0.87, confirming good separation but limited buyer detection.

Training Data

Fig 66:Mobile-Confusion Matrix and Performance Check



Observations:

1. Confusion matrix shows 4133 non-buyers correctly predicted, 111 non-buyers wrongly predicted, 282 buyers correctly predicted, and 496 buyers wrongly predicted.
2. Accuracy of the model is 87.91 percent, showing overall good performance.
3. Recall is 36.26 percent, meaning many actual buyers are missed.
4. Precision is 71.76 percent, meaning predicted buyers are mostly correct.
5. F1 score is 0.48 and ROC score is 0.89, showing good class separation but weak buyer capture.

For Laptop

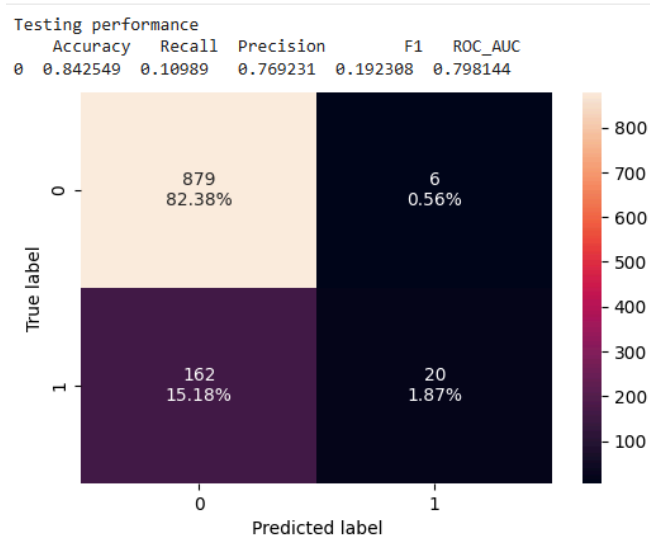
```
AdaBoostClassifier
AdaBoostClassifier(random_state=1)
```

Fig 67:Laptop-AdaBoost Classifier Model

Confusion Matrix and Performance Check of Testing data - For Laptop

Observations:

Fig 68:Laptop-Confusion Matrix and Performance Check

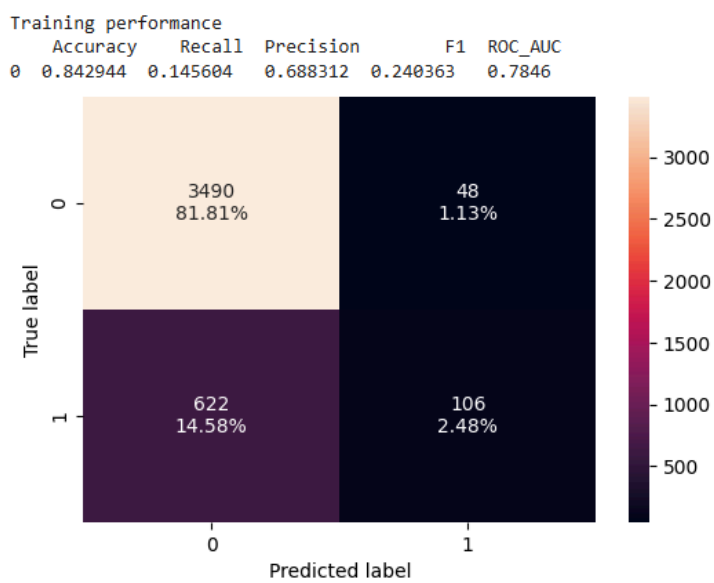


1. The confusion matrix shows 879 non-buyers correctly predicted and 20 buyers correctly predicted.
2. Accuracy is 84.25 percent, showing the model performs reasonably well on unseen data.
3. Recall is 10.99 percent, meaning most actual buyers are not identified by the model.
4. Precision is 76.92 percent, meaning when the model predicts a buyer, it is usually correct.
5. F1 score is 0.19 and ROC score is 0.80, indicating weak buyer detection but acceptable class separation.

Confusion Matrix and Performance Check of Training data - For Laptop

Observations:

Fig 69:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 3490 non-buyers correctly predicted and 106 buyers correctly predicted.
2. Accuracy is 84.29 percent, showing consistent performance on training data.
3. Recall is 14.56 percent, meaning the model still misses many actual buyers.
4. Precision is 68.83 percent, showing moderate correctness in buyer predictions.
5. F1 score is 0.24 and ROC score is 0.78, indicating limited buyer identification with fair separation ability.

AdaBoost With Hyperparameter Tuning - For Laptop

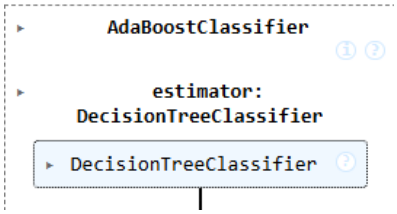
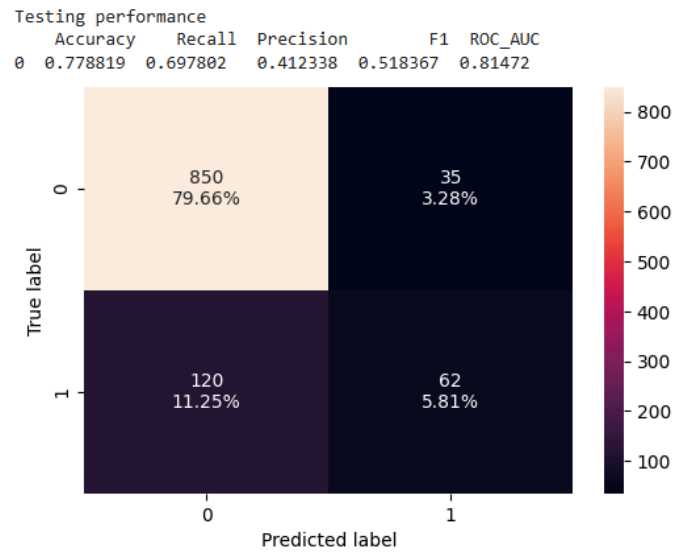


Fig 70:Laptop-Tuned AdaBoost Model

Confusion Matrix and Performance Check After Tuning - For Laptop

Testing Data

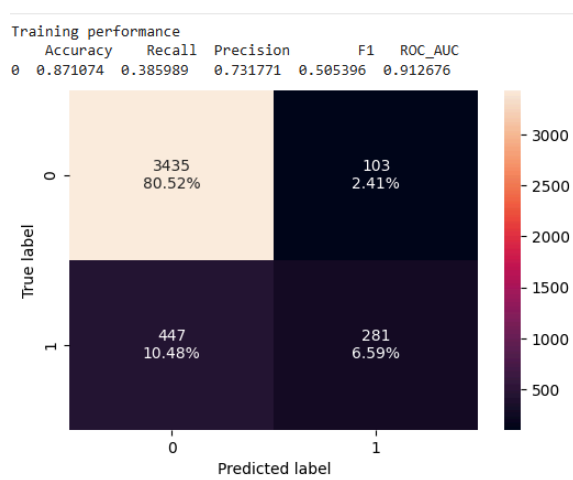
Fig 71:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 850 non-buyers correctly predicted and 62 buyers correctly predicted.
2. Accuracy is 77.88 percent, showing moderate overall prediction performance on test data.
3. Recall is 69.78 percent, meaning most actual ticket buyers are successfully identified.
4. Precision is 41.23 percent, meaning many predicted buyers are not actual buyers.
5. F1 score is 0.52 and ROC score is 0.81, indicating balanced buyer detection with good separation ability.

Training Data

Fig 72:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 3435 non-buyers correctly predicted and 281 buyers correctly predicted.
2. Accuracy is 87.11 percent, showing strong performance on training data.
3. Recall is 38.60 percent, meaning many actual buyers are missed during training.
4. Precision is 73.18 percent, meaning predicted buyers are mostly correct.
5. F1 score is 0.51 and ROC score is 0.91, indicating strong class separation with moderate buyer identification.

Gradient Boosting Classifier

For Mobile

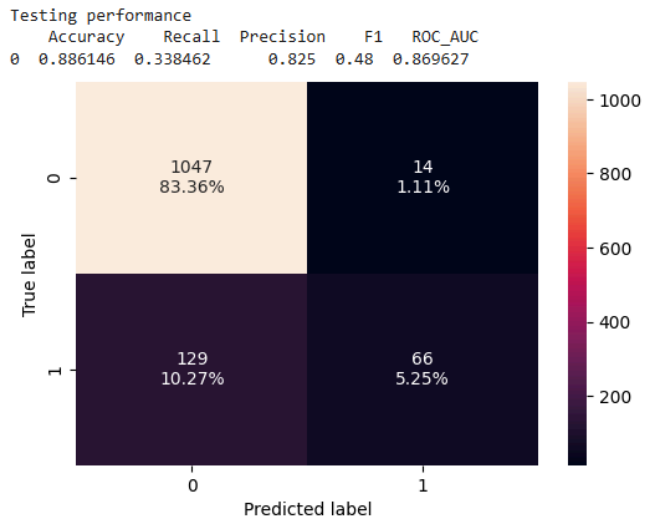
```
GradientBoostingClassifier
GradientBoostingClassifier(random_state=1)
```

Fig 73:Mobile-Gradient Boost Model

Confusion Matrix and Performance Check of Testing data - For Mobile

Observations:

Fig 74:Mobile-Confusion Matrix and Performance Check

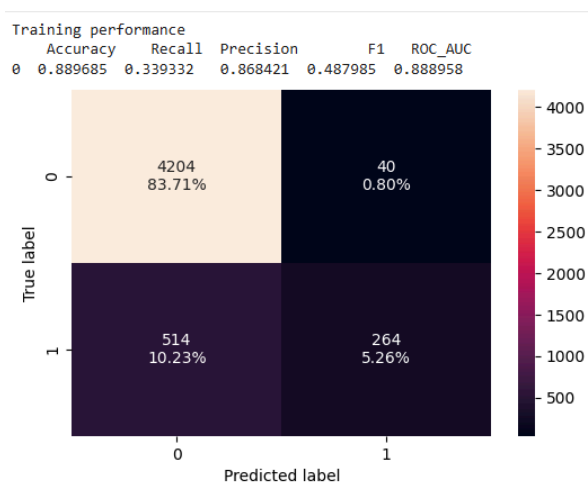


1. The confusion matrix shows 1047 non-buyers correctly predicted and 66 buyers correctly predicted.
2. Accuracy is 88.61 percent, showing strong overall prediction performance on test data.
3. Recall is 33.85 percent, meaning only one third of actual ticket buyers are identified.
4. Precision is 82.50 percent, meaning most predicted buyers are truly buyers.
5. F1 score is 0.48 and ROC score is 0.87, indicating good balance between buyer detection and class separation.

Confusion Matrix and Performance Check of Training data - For Mobile

Observations:

Fig 75:Mobile-Confusion Matrix and Performance Check



1. The confusion matrix shows 4204 non-buyers correctly predicted and 264 buyers correctly predicted.
2. Accuracy is 88.97 percent, showing consistent performance on training data.
3. Recall is 33.93 percent, meaning many actual buyers are still missed.
4. Precision is 86.84 percent, meaning predicted buyers are highly reliable.
5. F1 score is 0.49 and ROC score is 0.89, indicating strong class separation with moderate buyer identification.

Gradient Boosting With Hyperparameter Tuning - For Mobile

```
▸ GradientBoostingClassifier
▸ init: AdaBoostClassifier
  ▸ AdaBoostClassifier
```

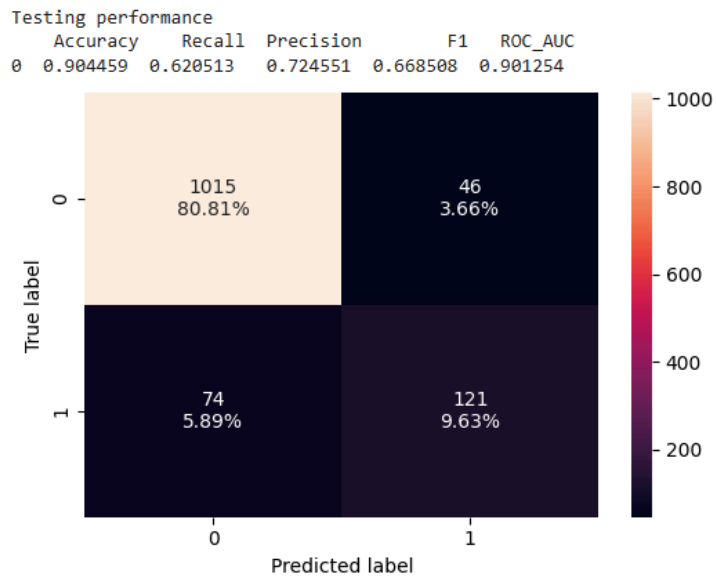
Fig 76:Mobile-Tuned Gradient Boost Model

Confusion Matrix and Performance Check After Tuning - For Mobile

Testing Data

1. The confusion matrix shows 1015 non-buyers correctly predicted and 121 buyers correctly predicted.
2. Accuracy is 90.45 percent, showing strong overall performance on unseen data.
3. Recall is 62.05 percent, meaning most actual ticket buyers are identified.
4. Precision is 72.46 percent, meaning predicted buyers are mostly correct.
5. F1 score is 0.67 and ROC score is 0.90, indicating good balance and strong ability to separate buyers and non-buyers.

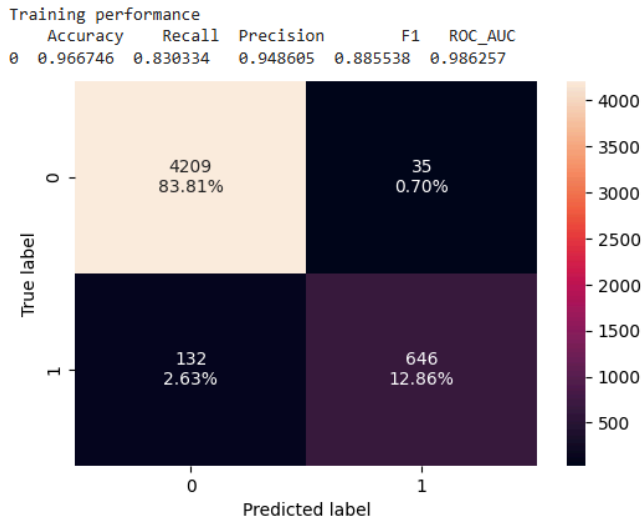
Fig 77:Mobile-Confusion Matrix and Performance Check



Training Data

1. The confusion matrix shows 4209 non-buyers correctly predicted and 646 buyers correctly predicted.
2. Accuracy is 96.67 percent, indicating excellent fit on training data.
3. Recall is 83.03 percent, meaning most actual buyers are captured during training.
4. Precision is 94.87 percent, meaning predicted buyers are highly reliable.

Fig 78:Mobile-Confusion Matrix and Performance Check



5. F1 score is 0.89 and ROC score is 0.99, showing very strong buyer identification and class separation.

For Laptop

```
GradientBoostingClassifier
GradientBoostingClassifier(random_state=1)
```

Fig 79:Laptop-Gradient Boost Model

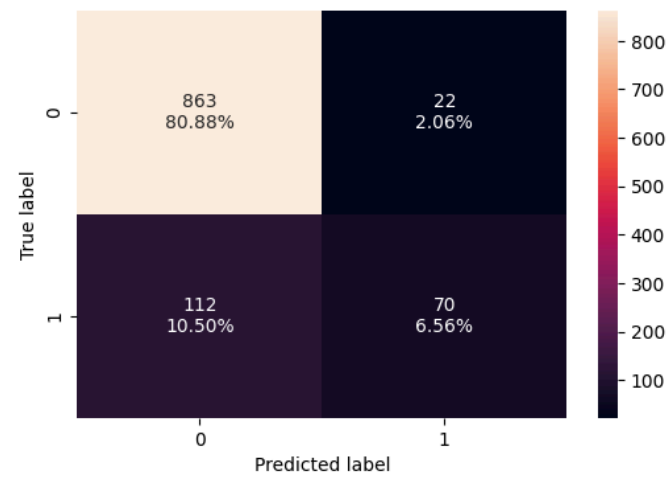
Confusion Matrix and Performance Check of Testing data - For Laptop

Observations:

1. The confusion matrix shows 863 non-buyers correctly predicted and 70 buyers correctly predicted.
2. Accuracy is 87.44 percent, showing good overall prediction on test data.
3. Recall is 38.46 percent, meaning many actual buyers are still missed.
4. Precision is 76.09 percent, meaning most predicted buyers are correct.
5. F1 score is 0.51 and ROC score is 0.87, indicating moderate balance and good class separation.

Fig 80:Laptop-Confusion Matrix and Performance Check

```
Testing performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.874414  0.384615    0.76087  0.510949  0.871326
```



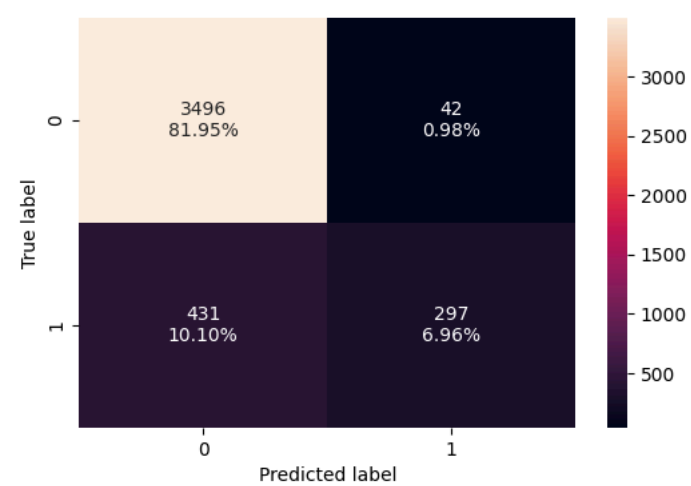
Confusion Matrix and Performance Check of Training data - For Laptop

Observations:

1. The confusion matrix shows 3496 non-buyers correctly predicted and 297 buyers correctly predicted.
2. Accuracy is 88.91 percent, indicating stable performance on training data.
3. Recall is 40.80 percent, showing limited ability to capture all buyers.
4. Precision is 87.61 percent, meaning predicted buyers are highly reliable.

Fig 81:Laptop-Confusion Matrix and Performance Check

```
Training performance
    Accuracy    Recall  Precision        F1   ROC_AUC
0  0.889123  0.407967   0.876106  0.556701  0.911869
```



5. F1 score is 0.56 and ROC score is 0.91, showing good separation between buyers and non-buyers.

Gradient Boosting With Hyperparameter Tuning - For Laptop

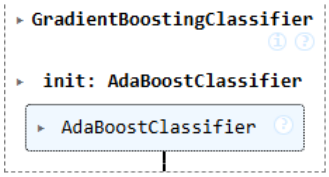
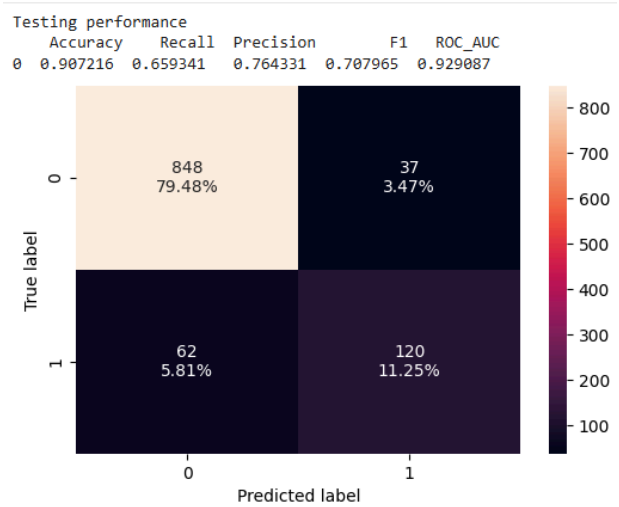


Fig 82:Laptop-Tuned Gradient Boost Model

Confusion Matrix and Performance Check After Tuning - For Laptop

Testing Data

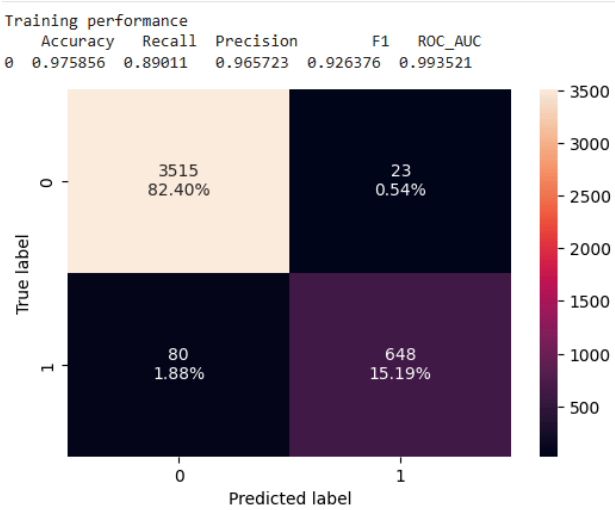
Fig 83:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 848 non-buyers correctly predicted and 120 buyers correctly predicted.
2. Accuracy is 90.72 percent, showing strong overall prediction on test data.
3. Recall is 65.93 percent, meaning the model identifies most actual ticket buyers.
4. Precision is 76.43 percent, meaning predicted buyers are mostly correct.
5. F1 score is 0.71 and ROC score is 0.93, indicating good balance and excellent separation between buyers and non-buyers.

Training Data

Fig 84:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 3515 non-buyers correctly predicted and 648 buyers correctly predicted.
2. Accuracy is 97.59 percent, showing very high performance on training data.
3. Recall is 89.01 percent, meaning most buyers are successfully captured.
4. Precision is 96.57 percent, meaning predicted buyers are highly accurate.

5. F1 score is 0.93 and ROC score is 0.99, indicating excellent model learning and class separation.

XGBoost Classifier

For Mobile

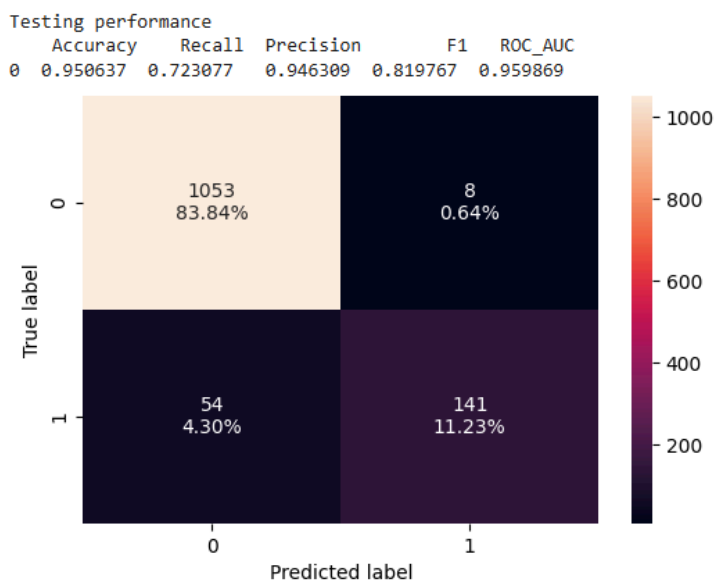
```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, feature_weights=None, gamma=None,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
```

Fig 85:Mobile- XGBoost Model

Confusion Matrix and Performance Check of Testing data - For Mobile

Observations:

Fig 86:Mobile-Confusion Matrix and Performance Check

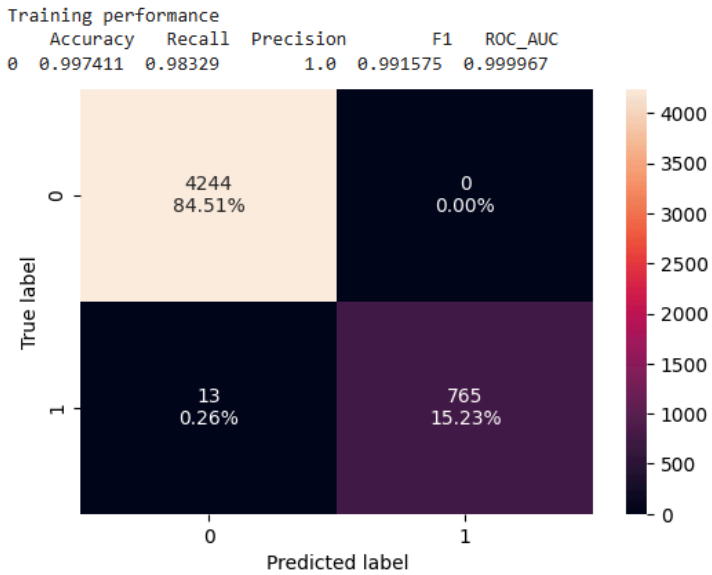


1. The confusion matrix shows 1053 non-buyers correctly predicted and 141 buyers correctly predicted.
2. Accuracy is 95.06 percent, showing very strong overall prediction on unseen mobile data.
3. Recall is 72.31 percent, meaning most actual mobile ticket buyers are identified, but some buyers are still missed.
4. Precision is 94.63 percent, meaning predicted buyers are mostly correct with very few false alarms.
5. F1 score is 0.82 and ROC score is 0.96, showing a good balance between precision and recall and excellent class separation.

Confusion Matrix and Performance Check of Training data - For Mobile

Observations:

Fig 87:Mobile-Confusion Matrix and Performance Check



1. The confusion matrix shows 4244 non-buyers correctly predicted and 765 buyers correctly predicted.
2. Accuracy is 99.74 percent, indicating extremely high performance on training data.
3. Recall is 98.33 percent, meaning almost all mobile ticket buyers are correctly captured.
4. Precision is 100 percent, meaning every predicted buyer is actually a buyer in training data.
5. F1 score is 0.99 and ROC score is 1.00, indicating near-perfect learning and a strong sign of overfitting risk.

XGBoost With Hyperparameter Tuning - For Mobile

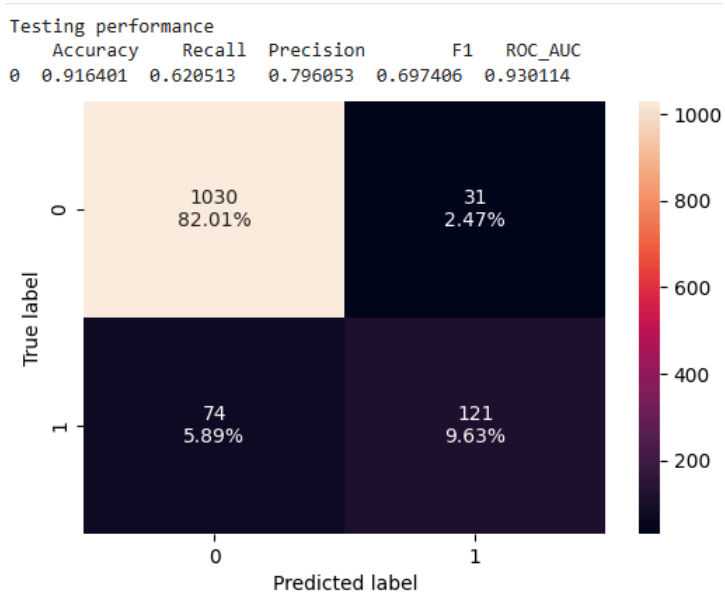
```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=0.6, colsample_bynode=None,
              colsample_bytree=0.8, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, feature_weights=None, gamma=3,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.2, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=100, n_jobs=None,
```

Fig 88:Mobile-Tuned XGBoost Model

Confusion Matrix and Performance Check After Tuning - For Mobile

Testing Data

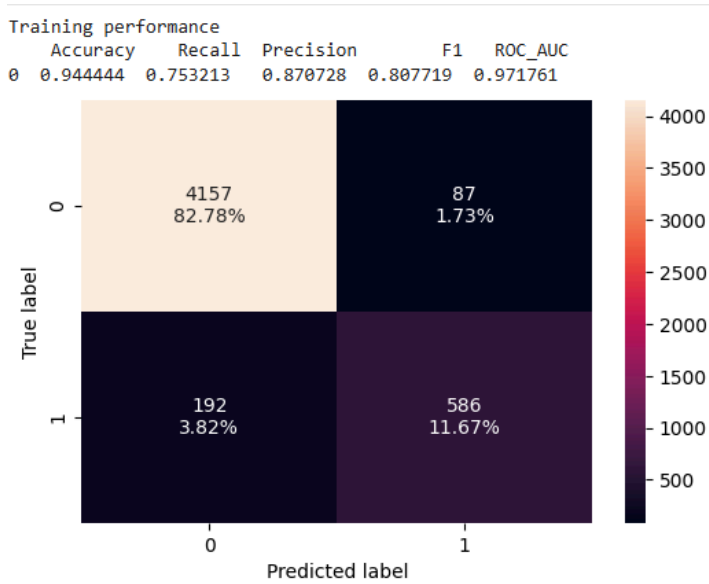
Fig 89:Mobile-Confusion Matrix and Performance Check



1. The confusion matrix shows 1030 non-buyers correctly predicted and 121 buyers correctly predicted.
2. Accuracy is 91.64 percent, showing strong overall performance on unseen mobile data.
3. Recall is 62.05 percent, meaning a moderate number of actual buyers are identified, but some buyers are still missed.
4. Precision is 79.61 percent, indicating most predicted buyers are correct with limited false predictions.
5. F1 score is 0.70 and ROC score is 0.93, showing a good balance between buyer detection and prediction reliability.

Training Data

Fig 90:Mobile-Confusion Matrix and Performance Check



1. The confusion matrix shows 4157 non-buyers correctly predicted and 586 buyers correctly predicted.
2. Accuracy is 94.44 percent, indicating very strong learning on training data.
3. Recall is 75.32 percent, meaning most buyers in training data are successfully identified.
4. Precision is 87.07 percent, showing high confidence when the model predicts a buyer.
5. F1 score is 0.81 and ROC score is 0.97, indicating excellent separation between buyers and non-buyers with controlled overfitting.

For Laptop

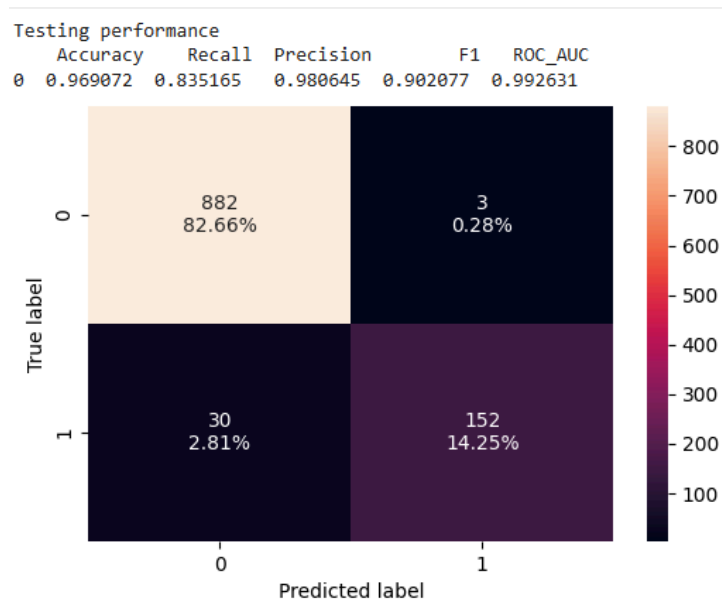
```
XGBClassifier
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, feature_weights=None, gamma=None,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
              num_parallel_tree=None, ...)
```

Fig 91:Laptop- XGBoost Model

Confusion Matrix and Performance Check of Testing data - For Laptop

Observations:

Fig 92:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 882 non-buyers correctly predicted and 152 buyers correctly predicted.
2. Accuracy is 96.91 percent, showing excellent overall performance on test data.
3. Recall is 83.52 percent, meaning most actual ticket buyers are identified correctly.
4. Precision is 98.06 percent, meaning predicted buyers are almost always correct.
5. F1 score is 0.90 and ROC score is 0.99, indicating very strong balance and excellent class separation.

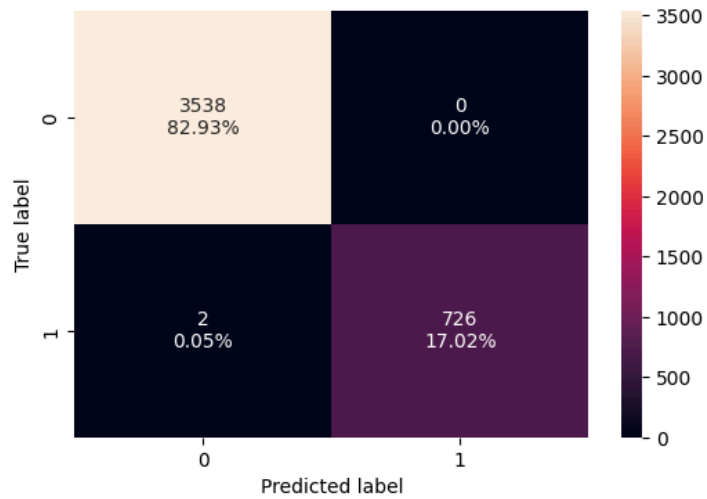
Confusion Matrix and Performance Check of Training data - For Laptop

Observations:

1. The confusion matrix shows 3538 non-buyers correctly predicted and 726 buyers correctly predicted.
2. Accuracy is 99.95 percent, showing near-perfect performance on training data.
3. Recall is 99.73 percent, meaning almost all buyers are captured.
4. Precision is 100 percent, meaning every predicted buyer is correct.
5. F1 score is 0.99 and ROC score is 1.00, indicating perfect learning and very high risk of overfitting.

Fig 93:Laptop-Confusion Matrix and Performance Check

```
Training performance
    Accuracy    Recall  Precision        F1  ROC_AUC
0  0.999531  0.997253        1.0  0.998624      1.0
```



XGBoost With Hyperparameter Tuning - For Laptop

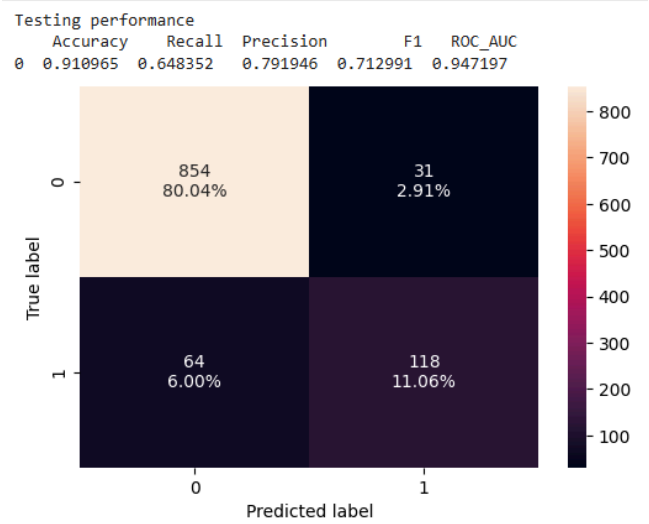
```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=0.8, colsample_bynode=None,
              colsample_bytree=0.8, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, feature_weights=None, gamma=3,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.2, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=100, n_jobs=None,
```

Fig 94:Laptop-Tuned XGBoost Model

Confusion Matrix and Performance Check After Tuning - For Mobile

Testing Data

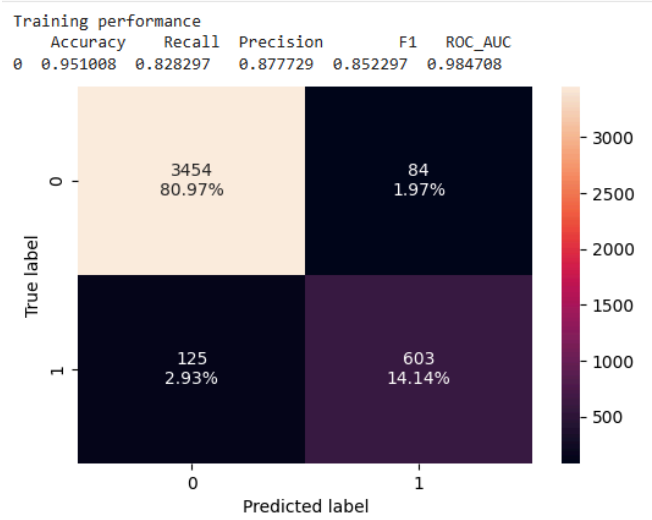
Fig 95:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 854 non-buyers correctly predicted and 118 buyers correctly predicted.
2. Accuracy is 91.10 percent, showing strong overall prediction performance on new laptop data.
3. Recall is 64.84 percent, meaning the model captures a fair number of actual laptop ticket buyers but still misses some.
4. Precision is 79.19 percent, indicating that most predicted buyers are truly buyers with limited false alarms.
5. F1 score is 0.71 and ROC score is 0.95, showing a good balance between buyer detection and class separation.

Training Data

Fig 96:Laptop-Confusion Matrix and Performance Check



1. The confusion matrix shows 3454 non-buyers correctly predicted and 603 buyers correctly predicted.
2. Accuracy is 95.10 percent, indicating excellent learning on the training data.
3. Recall is 82.83 percent, meaning most actual laptop ticket buyers are correctly identified.
4. Precision is 87.77 percent, showing high confidence when predicting a buyer.
5. F1 score is 0.85 and ROC score is 0.98, indicating strong model stability and minimal overfitting.

Model Comparison and Final Model Selection

The Model Performance Comparison for Testing Data Set - For Mobile

The Model Performance Comparison for Testing Data Set - For Mobile:

	Decision Tree mobile	Tuned Decision Tree mobile	Bagging mobile	Tuned Bagging mobile	Random Forest mobile	Tuned Random Forest mobile	AdaBoost mobile	Tuned AdaBoost mobile	Gradient Boosting mobile	Tuned Gradient Boosting mobile	XGBoost mobile	Tuned XGBoost mobile
Accuracy	0.928344	0.790605	0.948248	0.960987	0.960987	0.771497	0.863057	0.872611	0.886146	0.904459	0.950637	0.916401
Recall	0.733333	0.625641	0.702564	0.769231	0.748718	0.702564	0.189744	0.379487	0.338462	0.620513	0.723077	0.620513
Precision	0.790055	0.391026	0.951389	0.974026	1.000000	0.374317	0.725490	0.654867	0.825000	0.724551	0.946309	0.796053
F1	0.760638	0.481262	0.808260	0.859599	0.856305	0.488414	0.300813	0.480519	0.480000	0.668508	0.819767	0.697406
ROC_AUC	0.848759	0.787612	0.963423	0.988407	0.986597	0.827154	0.817468	0.870947	0.869627	0.901254	0.959869	0.930114

Fig 97: The Model Performance Comparison for Testing Data Set - For Mobile

The Model Performance Comparison for Training Data Set - For Mobile

The Model Performance Comparison for Training Data Set - For Mobile:

	Decision Tree mobile	Tuned Decision Tree mobile	Bagging mobile	Tuned Bagging mobile	Random Forest mobile	Tuned Random Forest mobile	AdaBoost mobile	Tuned AdaBoost mobile	Gradient Boosting mobile	Tuned Gradient Boosting mobile	XGBoost mobile	Tuned XGBoost mobile
Accuracy	1.0	0.793907	0.995221	1.0	1.0	0.762644	0.851254	0.879132	0.889685	0.966746	0.997411	0.944444
Recall	1.0	0.632391	0.970437	1.0	1.0	0.695373	0.133676	0.362468	0.339332	0.830334	0.983290	0.753213
Precision	1.0	0.396454	0.998677	1.0	1.0	0.361631	0.587571	0.717557	0.868421	0.948605	1.000000	0.870728
F1	1.0	0.487370	0.984355	1.0	1.0	0.475814	0.217801	0.481640	0.487985	0.885538	0.991575	0.807719
ROC_AUC	1.0	0.783525	0.999959	1.0	1.0	0.813568	0.787507	0.892276	0.888958	0.986257	0.999967	0.971761

Fig 98: The Model Performance Comparison for Training Data Set - For Mobile

The Model Performance Comparison for Testing Data Set - For Laptop

The Model Performance Comparison for Testing Data Set - For Laptop:

	Decision Tree laptop	Tuned Decision Tree laptop	Bagging laptop	Tuned Bagging laptop	Random Forest laptop	Tuned Random Forest laptop	AdaBoost laptop	Tuned AdaBoost laptop	Gradient Boosting laptop	Tuned Gradient Boosting laptop	XGBoost laptop	Tuned XGBoost laptop
Accuracy	0.938144	0.669166	0.938144	0.969072	0.961575	0.778819	0.842549	0.854733	0.874414	0.907216	0.969072	0.910965
Recall	0.835165	0.785714	0.835165	0.846154	0.785714	0.697802	0.109890	0.340659	0.384615	0.659341	0.835165	0.648352
Precision	0.808511	0.312910	0.808511	0.968553	0.986207	0.412338	0.769231	0.639175	0.760870	0.764331	0.980645	0.791946
F1	0.821622	0.447574	0.821622	0.903226	0.874618	0.518367	0.192308	0.444444	0.510949	0.707965	0.902077	0.712991
ROC_AUC	0.897243	0.758872	0.897243	0.994102	0.993956	0.814720	0.798144	0.871391	0.871326	0.929087	0.992631	0.947197

Fig 99: The Model Performance Comparison for Testing Data Set - For Laptop

The Model Performance Comparison for Training Data Set - For Laptop

The Model Performance Comparison for Training Data Set - For Laptop:

	Decision Tree laptop	Tuned Decision Tree laptop	Bagging laptop	Tuned Bagging laptop	Random Forest laptop	Tuned Random Forest laptop	AdaBoost laptop	Tuned AdaBoost laptop	Gradient Boosting laptop	Tuned Gradient Boosting laptop	XGBoost laptop	Tuned XGBoost laptop
Accuracy	1.0	0.686357	1.0	1.0	1.0	0.772621	0.842944	0.871074	0.889123	0.975856	0.999531	0.951008
Recall	1.0	0.810440	1.0	1.0	1.0	0.697802	0.145604	0.385989	0.407967	0.890110	0.997253	0.828297
Precision	1.0	0.329609	1.0	1.0	1.0	0.403816	0.688312	0.731771	0.876106	0.965723	1.000000	0.877729
F1	1.0	0.468626	1.0	1.0	1.0	0.511581	0.240363	0.505396	0.556701	0.926376	0.998624	0.852297
ROC_AUC	1.0	0.797004	1.0	1.0	1.0	0.823464	0.784600	0.912676	0.911869	0.993521	1.000000	0.984708

Fig 100: The Model Performance Comparison for Training Data Set - For Laptop

Observations:

1. Tuned XGBoost shows the best balance of accuracy, recall, F1 score, and ROC AUC on test data for both mobile and laptop.
2. Other models like Decision Tree and AdaBoost perform well on training data but drop on test data, which shows overfitting.
3. Confusion matrices of Tuned XGBoost show fewer false predictions compared to other models.
4. For mobile, Tuned XGBoost has good precision and acceptable recall, meaning most predicted buyers are correct.
5. For laptop, Tuned XGBoost has higher recall and F1 score than mobile, meaning it identifies buyers better.
6. Laptop users purchase tickets more than mobile users, as seen from higher true positive counts in confusion matrices.
7. The final model selected is Tuned XGBoost for both mobile and laptop due to stable and reliable test performance.

Overall Insights & Recommendations

Insights

1. Laptop users are more likely to complete ticket purchases than mobile users, as models consistently identify more buyers on laptop data.
2. XGBoost and tuned Gradient Boosting show strong ability to separate buyers and non-buyers, making them reliable for prediction.
3. Recall is lower in simpler models, meaning many potential buyers are missed without advanced models.
4. High training performance in some models indicates overfitting, so test performance should be prioritized for decisions.
5. Confusion matrices show that tuned models reduce missed buyers, which is critical for revenue growth.

Recommendations

1. Focus marketing campaigns more on laptop users, as they show higher purchase completion rates.
2. Use XGBoost as the final prediction model to identify high-intent customers accurately.
3. For mobile users, improve user experience and checkout flow to reduce drop-offs and increase conversions.
4. Target predicted buyers with personalized offers, reminders, or discounts to maximize ticket sales.
5. Continuously monitor model performance on new data to avoid overfitting and maintain accuracy over time.